

Micoprism-based two-photon imaging of the mouse inferior colliculus reveals novel organizational principles of the auditory midbrain

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Abstract

To navigate real-world listening conditions, the auditory system relies on the integration of multiple sources of information. However, to avoid inappropriate cross-talk between inputs, highly connected neural systems need to strike a balance between integration and segregation. Here, we develop a novel approach to examine how repeated neurochemical modules in the mouse inferior colliculus lateral cortex (LC) allow controlled integration of its multimodal inputs. The LC had been impossible to study via imaging because it is buried in a sulcus. Therefore we coupled two-photon microscopy with the use of a micoprism to reveal the first-ever sagittal views of the LC to examine neuronal responses with respect to its neurochemical motifs under anesthetized and awake conditions. This approach revealed marked differences in the acoustic response properties of LC and neighboring nonlemniscal portions of the inferior colliculus. In addition, we observed that the module and matrix cellular motifs of the LC displayed distinct somatosensory and auditory responses. Specifically, neurons in modules demonstrated primarily offset responses to acoustic stimuli with enhancement in responses to bimodal stimuli, whereas matrix neurons showed onset response to acoustic stimuli and suppressed responses to bimodal stimulation. Thus, this new approach revealed that the repeated structural motifs of the LC permit functional integration of multimodal inputs while retaining distinct response properties.

eLife assessment

This study provides **valuable** new insights into how multisensory information is processed in the lateral cortex of the inferior colliculus, a poorly understood part of the auditory midbrain. By developing new imaging techniques that provide the first optical access to the lateral cortex in a living animal, the authors provide **convincing** in vivo evidence that this region contains separate subregions that can be distinguished by their sensory inputs and neurochemical profiles, as suggested by previous anatomical and in vitro studies. This work provides a foundation for future research exploring how this part of the auditory midbrain contributes to multisensory-based behavior.

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Introduction

Real-world listening occurs in environments cluttered by noise. To navigate these conditions, the auditory system relies on the integration of multiple sources of input. Broad integration of neural inputs, however, comes at a potential cost. Dense convergence can lead to inappropriate cross-talk between input streams, generating a processing “soup” whereby individual neural pathways cannot be independently modulated. Therefore, highly connected neural systems need to preserve the specificity of inputs in the face of massive integration (Fair et al., 2007 [↗](#); Sporns, Tononi, & Edelman, 2000 [↗](#); Tononi, Sporns, & Edelman, 1994 [↗](#))

Several brain systems have solved the integration/segregation problem by creating repeated structural motifs that allow independent modulation of different pathways. For example, in the basal ganglia, neurochemical periodicity of the striosome and matrix regions permits the retention of functional independence of input streams in the face of massive convergence. As a result, basal ganglia direct and indirect pathways may be independently modulated, while allowing controlled crosstalk between channels (Alexander, Crutcher, & DeLong, 1990 [↗](#); Brimblecombe & Cragg, 2017 [↗](#); Draganski et al., 2008 [↗](#); Gerfen, 1985 [↗](#))

In the auditory system, the integration/segregation challenge is particularly formidable in the inferior colliculus (IC). The IC serves as a specialized integration hub because of its heavy innervation by higher auditory centers as well as by nonauditory regions. Thus, the IC stands at the crossroads between multiple brain regions and is thus important for linking acoustic stimuli with non-acoustic inputs. The IC comprises a lemniscal division, the central nucleus (ICC), that receives primarily ascending auditory projections (Oliver, 2005; Willard & Ryugo, 1983 [↗](#); Willott, 2001 [↗](#)) and two non-lemniscal divisions known as the dorsal (DC) and lateral (LC) cortices that receive massive descending auditory as well as non-auditory input (Bajo & Moore, 2005 [↗](#); Coleman & Clerici, 1987 [↗](#); Loftus, Malmierca, Bishop, & Oliver, 2008 [↗](#); Saldaña, Feliciano, & Mugnaini, 1996 [↗](#); Schreiner & Winer, 2005 [↗](#); Winer, Larue, Diehl, & Hefti, 1998 [↗](#)) as reviewed in (A. M. H. Lesicko & Geffen, 2022 [↗](#)).

The LC and DC appear to differ in their strategy for integrating these inputs. The DC does not contain known structural heterogeneities but has a tonotopic arrangement of neuronal responses on its surface (Barnstedt, Keating, Weissenberger, King, & Dahmen, 2015 [↗](#); A. B. Wong & Borst, 2019 [↗](#)). In contrast, the LC contains repeated neurochemical motifs, here referred to as modules, that express the inhibitory neurotransmitter GABA as well as other metabolic markers (Chernock, Larue, & Winer, 2004 [↗](#)). Areas outside the modules, referred to as matrix, stain strongly for

calretinin (Dillingham, Gay, Behrooz, & Gabriele, 2017 [↗](#)). We and others have determined that nearly all of the long-and short-range connectivity of the LC is governed by the module/matrix organization (Dillingham et al., 2017 [↗](#); Lamb-Echegaray, Noftz, Stinson, & Gabriele, 2019 [↗](#); A. M. H. Lesicko, Sons, & Llano, 2020 [↗](#); A. M. Lesicko, Hristova, Maigler, & Llano, 2016 [↗](#)).

Unfortunately, how the LC integrates these inputs while maintaining distinct paths of information flow is not known. For example, LC neurons respond to multisensory stimuli (Jain & Shore, 2006 [↗](#); Zhou & Shore, 2006 [↗](#)), but information from different modalities enters the LC in segregated channels (A. M. Lesicko et al., 2016 [↗](#)). Thus, integration across the module/matrix border must occur, but the mechanisms of this integration are unknown. Traditional methods of studying the LC, such as in vivo electrophysiology, pose significant challenges in unambiguously assigning individually recorded neurons to residence in matrix or modules.

Imaging approaches, such as two-photon (2P) microscopy, have proven to be enormously valuable in understanding the fine spatial structure of neural response properties in the auditory system (Bandyopadhyay, Shamma, & Kanold, 2010 [↗](#); Romero et al., 2020 [↗](#); Rothschild, Nelken, & Mizrahi, 2010 [↗](#)). However, because of its superficial location on the dorsal surface of the mouse midbrain, the DC is the only part of the IC that has been characterized by 2P imaging (Barnstedt et al., 2015 [↗](#); Ito, Hirose, Murase, & Ikeda, 2014 [↗](#); A. B. Wong & Borst, 2019 [↗](#)). Optically, it has been impossible to characterize the functions of the LC, which is laterally embedded deep in a brain sulcus.

Here, we describe a novel approach using a 45° micropism inserted into the sulcus that separates the LC from the cerebellum to investigate the fine-scale functional organization of the LC from its lateral surface through 2P imaging. The 2P images obtained from the micropism revealed the first-ever images of the LC in vivo showing its characteristic GABAergic modules. Compared to the DC imaged from the surface, the acoustically and somatosensory-evoked calcium signals obtained by the 2P imaging of the LC through the micropism revealed functional distinctions between the different cellular motifs of the LC. For instance, while modules were more responsive to somatosensory stimulation than matrix, the latter was more acoustically-responsive and had a lower detection threshold for complex sounds. Additionally, modules and matrix were found to differently process the spectral and temporal features of the acoustic information. These findings suggest that the non-lemniscal divisions of the IC contain rich but distinct platforms for the processing of different features of auditory and non-auditory information.

Results

The insertion of 45° micropism for imaging the LC from the side

GAD67-GFP knock-in mice on a Swiss Webster background were crossed with Thy1-jRGECO1a mice to produce a GAD67-GFPx jRGECO1a hybrid mouse, which was used to visualize GABAergic modules, to identify the dense network of GABAergic neurons and terminals in modules of the LC in the green channel (Tamamaki et al., 2003 [↗](#)), and the calcium indicator, jRGECO1a, in the red channel (Dana et al., 2018 [↗](#)). A standard craniotomy was performed (Goldey et al., 2014 [↗](#)) with modifications to suit the specific characteristics of the IC in terms of its location, depth, and high density of the surrounding vasculature (Xiong et al., 2017 [↗](#)). The craniotomy (~ 3 x 4 mm) did not drill directly into the sagittal and lambdoid sutures which allowed minimal disruption of the superficial sinuses below the skull (**Figure 1A** [↗](#)) as indicated by the minimal bleeding near the bone flap (**Figure 1B** [↗](#)). During drilling, pressurized air and a cold stream of saline were used to remove the bone debris and dissipate heat. Because the IC sits deeper in the brain compared to the surrounding structures, the direct insertion of the 45° micropism was found to push the IC ventrally. Therefore, a fine diamond knife was used to separate the lateral surface of the IC from the surrounding tissues without damaging the transverse and sigmoid sinuses (**Figures 1C and**

D [↗](#)). This step was followed by a gentle retraction of the knife medially to generate a pocket to secure the microprism (**Figure 1E** [↗](#)). The successful insertion of the microprism was assessed by examining the reflection of the diamond knife from the microprism's top glass surface (**Figures 1F and G** [↗](#)). Holding the microprism down and retracting the knife upward simultaneously resulted in securing the microprism in its pocket (**Figure 1H and I** [↗](#)). Locking the microprism down in its pocket for 10 minutes with the diamond knife (**Figure 1J** [↗](#)) is critical to ensure that the microprism is stabilized and to prevent bleeding. Filling the gap between the skull surface and the microprism with saline was found to generate motion artifact due to the heart pulsation, so 1% agarose was used to fill the gap to reduce the motion artifact and add more stability to the microprism (**Figure 1K** [↗](#)). To compare response properties between the LC and DC, direct imaging (i.e., without a microprism) of the dorsal surface of the IC, where the DC is located, was also conducted by scanning the DC surface from the most medial to lateral regions following the convex curvature of the surface (**Figures 1L and M** [↗](#)).

The 2P characterization of GABAergic and non-GABAergic neuronal distribution on the LC and DC surfaces with histological validation

Following the surgery, the anesthetized animals were taken immediately for imaging. Examination of the distribution of GABAergic modules in the LC was critical to validate the target area for functional calcium imaging. Therefore, the 920 nm laser beam was initially used to scan for GFP, which is expressed in the GABAergic cells of GAD67-GFP knock-in mice. The insertion of the microprism was initially viewed by low magnification fluorescence imaging of the GFP signals from the whole surface of the IC along with the inserted microprism (**Figure 2A** [↗](#)). This step validated the location of the microprism relative to the IC under the microscope and ensured there was no bleeding that could block the field of view. The 2P images of the GFP signals from the LC imaged via the microprism, or for simplicity LC_(microprism), across different animals revealed the LC characteristic GABAergic modules (**Figures 2C and E** [↗](#), the irregular white lines), which were consistent with previous 3D reconstructions of modules from the coronal sections (Chernock et al., 2004 [↗](#); A. M. Lesicko et al., 2016 [↗](#)). To further validate the microprism location and the ability to image the LC from the lateral surface, a laser lesion (see Methods) was created via the microprism (**Figure 2G** [↗](#)), which was validated histologically (**Figure 2H** [↗](#)). For comparison, the whole dorsal surface of the IC was scanned from medial to lateral borders (**Figure 2B** [↗](#), the dotted white lines). Unlike the LC, most of the dorsal surface of the IC showed a non-clustered distribution of the GABAergic cells (**Figures 2D and F** [↗](#), blue boxes). However, the lateral horizon of the IC dorsal surface showed clusters of GABAergic cells and terminals that were confirmed to be modules, which comprise the prominent feature of the LC (**Figures 2D and F** [↗](#), red boxes). To further validate the presence of GABAergic modules in the dorsolateral surface of the IC, a laser lesion was created in a GABAergic module located in that area (**Figures 2I and J** [↗](#), dotted red circle), which was confirmed histologically (**Figure 2K** [↗](#)), demonstrating that it is possible to image a portion of the LC directly from its dorsal surface.

Using the 1040 nm laser, the GAD67-GFPx jRGECO1a hybrid mouse showed a non-clustered cellular expression of jRGECO1a on the surface of the DC (**Figure 3A** [↗](#)) or on the surface of the LC_(microprism) (**Figure 3B** [↗](#)), as expected. The IC of anesthetized animals in each experimental setup was then imaged to measure calcium signals over the course of pseudorandom presentations of either 500 ms pure tones of different combinations of frequencies (5–40 kHz, half octave intervals) and sound levels (40–80 dB SPL, 10 dB intervals) or broadband noise without or with amplitude modulation (AM-noise) of different combinations of AM frequencies (2–256 Hz, one-octave intervals) and sound levels (40–80 dB SPL, 10 dB intervals). As shown by two representative cells from the DC imaged from the dorsal surface or LC imaged via microprism, these acoustic stimulation paradigms successfully elicited transient cellular calcium signals that varied depending on the stimulus type (**Figures 3C and D** [↗](#)). The expression distributions of jRGECO1a were histologically examined for the 2P imaged areas of the DC and LC (**Figure 3** [↗](#) -

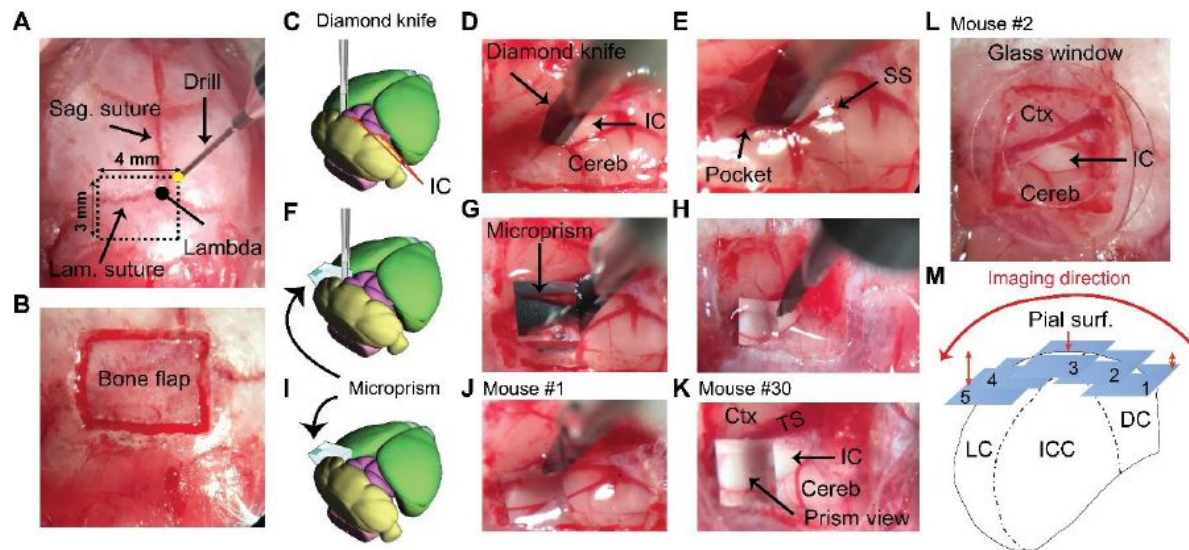


Figure 1

A brief Illustration of the surgical procedures.

(A) A surface view of the mouse's skull showing the landmarks of the craniotomy over the IC. (B) An image of the bone flap after finishing the drilling. (C-E) A cartoon and two images of the brain surface showing the insertion and the pocket made by the diamond knife at the lateral surface of the left IC. (F-H) A cartoon and two images of the brain surface show the insertion of the microprism into its pocket to directly face the LC after displacing the blade of the diamond knife. (I-K) A cartoon and two images of the brain surface show the final setup of the microprism for two different animals. (L) An image of the brain surface shows the placement of a glass window for the surface imaging of the DC. (M) A cartoon of the coronal section of the IC demonstrating the different fields of views taken along the mediolateral axis parallel to the curvature of the IC surface at different depths (1-5) from the pial surface. Cereb: Cerebellum, Ctx: Cerebral cortex, DC: Dorsal cortex of the IC, IC: Inferior colliculus, ICC: Central nucleus of the IC, Lam. Lambdoid, LC: Lateral cortex of the IC, Sag. Sagittal, SS: Sigmoid sinus, Surf: Surface, TS: Transverse sinus.

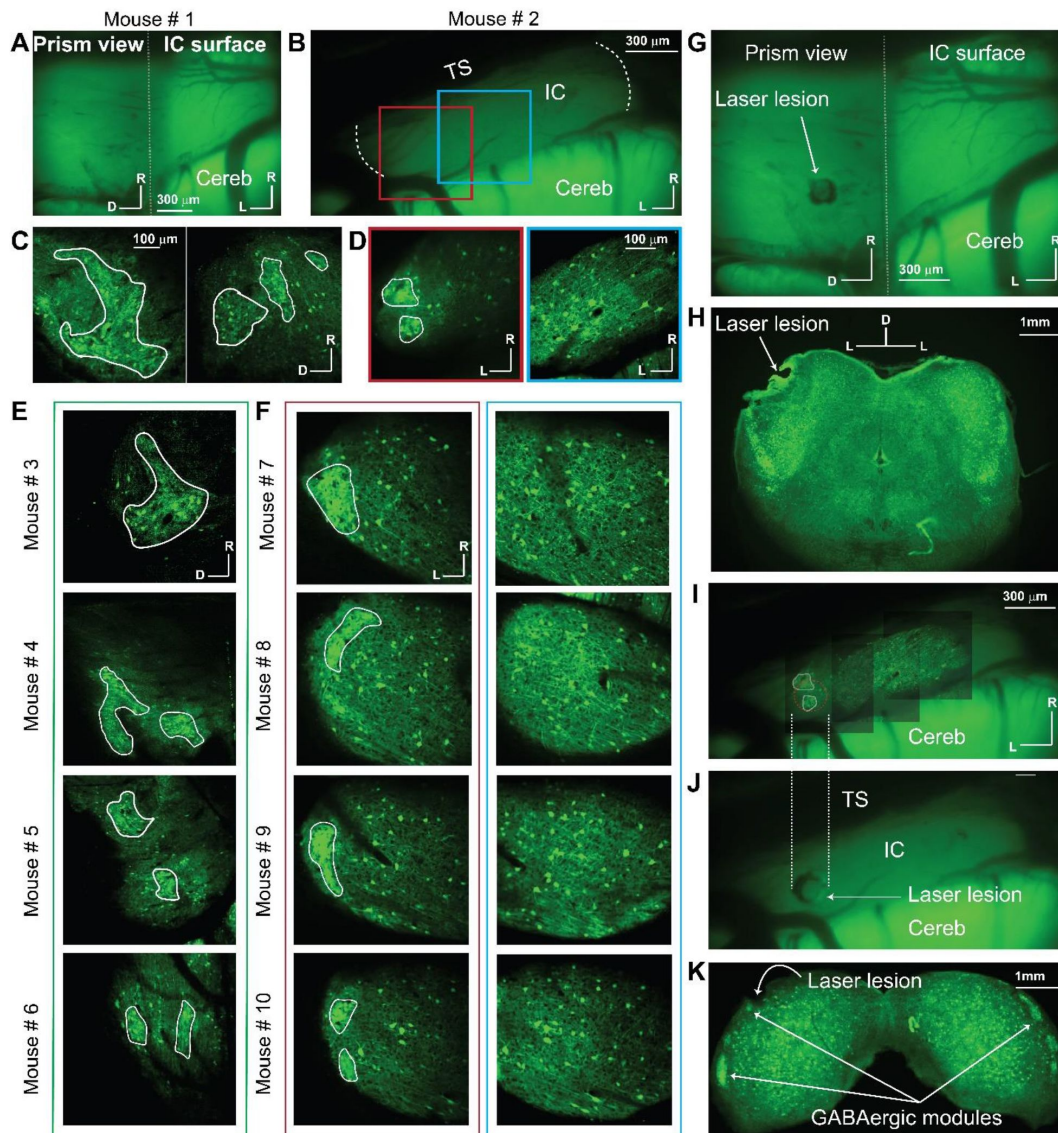


Figure 2

The distribution of GABAergic cells on the LC_(microprism) and DC imaged from the dorsal surface.

All images were taken for the GFP signals exclusively expressed in GABAergic cells. (A-B) Low magnification of fluorescence images for the GFP signals obtained from the IC surface with or without the microprism, respectively. The dotted white line in B represented the medial and lateral horizons of the IC. (C and E) The 2P images of the GABAergic cells on the LC_(microprism) show the GABAergic modules across multiple animals. (D and F) The 2P images of the GABAergic cells on either the dorsolateral surface of the DC imaged directly from its dorsal surface showing the GABAergic modules across multiple animals (red boxes) or its dorsomedial surface showing a homogenous distribution of GABAergic cells of different sizes with no signs of the GABAergic modules across multiple animals (blue boxes). (G) A low magnification fluorescence image of the IC surface with the microprism showing a laser lesion spot made through the microprism to validate the position of the microprism relative to the IC. (H) A histological coronal section at the level of the IC shows that the laser lesion made by the microprism was located at the sagittal surface of the LC_(microprism). (I-J) Low-magnification fluorescence images of the IC surface from the dorsal surface show a laser lesion spot made on the dorsolateral surface of the LC on the GABAergic module. The dotted white lines represent the alignment of the laser lesion (dotted red circle) to the location of the GABAergic module at the dorsolateral surface. (K) A histological coronal section at the level of the IC shows that the laser lesion made on the IC surface targeted the GABAergic module at the dorsolateral portion of the LC. All solid irregular white lines were made at the border of the GABAergic modules; D: Dorsal, IC: Inferior colliculus, L: Lateral, R: Rostral. Cereb: Cerebellum, TS: Transverse sinus.

figure supplement 1A). The DC and LC, have a similar fraction of cells expressing jRGECO1a (paired sample t-Test, $t_{(9)} = 0.98$, $p = 0.34$, $n = 10$ coronal sections from 2 animals) (Figure 3-figure supplement 1B and C). Also, the cells expressing jRGECO1a in all regions of the IC were only nonGABAergic neurons as they did not express GFP which is expressed exclusively in the GABAergic neurons (Figure 3-figure supplement 1D).

Acoustic responsiveness differs in the LC and DC

As indicated by the GFP signals and consistent with Figure 2, imaging the LC of the hybrid mice via microprism showed clusters of GABAergic modules, which take different irregular shapes on the surface of the LC (Figure 4A - irregular white lines). Based on the best pure tone frequency (BTF) of each responsive cell (Barnstedt et al., 2015), an inconsistent and fragmentary tonotopic organization was seen over the LC_(microprism) (Figure 4B) as indicated by a low R^2 of the best linear and quadratic regressions fit made between the BTFs of cells and their locations (Figure 4-figure supplement 1A). Additionally, the best fit of these regressions was found to be along different axes across the tested animals (Figure 4-figure supplement 1A). Despite this inconsistent topographical organization, it was found that the cells in the modules were tuned to relatively higher frequencies than those in the matrix (Mann-Whitney Test: $z = 4.03$, $p = 5.6 \times 10^{-5}$, a median of best-tuned frequency = 10 and 7.1 Hz for modules (215 cells from 7 animals) and matrix, (230 cells from 7 animals) respectively).

For comparison, the surface of the DC was imaged directly from the dorsal surface using the routine imaging setup. To view the majority of the DC surface, imaging was done by scanning the DC from its medial to the lateral horizon to generate a composite map (see Methods). As indicated by the GFP signals, it was noted that the most lateral region of the surface of the IC showed clusters of the GABAergic cells and terminals (i.e., modules, Figure 4C – irregular white lines) and are therefore considered part of the LC. Based on the BTF of each responsive cell (Barnstedt et al., 2015), the DC showed a tonotopic organization of cells along the rostromedial to caudolateral axis tuned to lower or higher frequencies, respectively, (Figure 4D) as indicated by a high R^2 of the regression fits made between BTFs of cells and their locations along the rostromedial to caudolateral axis. (Figure 4-figure supplement 1B). It was noted that the R^2 of the nonlinear quadratic regression was higher than that of the linear fit, which allowed that model to include the cells tuned to higher frequencies at the rostromedial part of the DC, which is consistent with the tonotopic organization that was previously described (Barnstedt et al., 2015; A. B. Wong & Borst, 2019). It was also noticed that a large area of the DC was tuned to the lowest tested frequency (5kHz), suggesting a low-frequency bias on the surface of the DC. To quantify tuning differences between DC and LC, the cumulative distribution function based on the cells' BTFs showed a low-frequency bias in the DC, similar to previous work (Ito et al., 2014; Romand & Ehret, 1990; Shen, Xu, & Shen, 2003; Stiebler & Ehret, 1985). Although around 50% of the cells of LC_(microprism) showed also a low-frequency bias to 5 and 7.1 kHz, collectively, most of the LC cells were tuned to relatively higher frequencies than those of the DC (Figure 5A, Mann-Whitney Test: $z = -11.34$, $p < 0.001$).

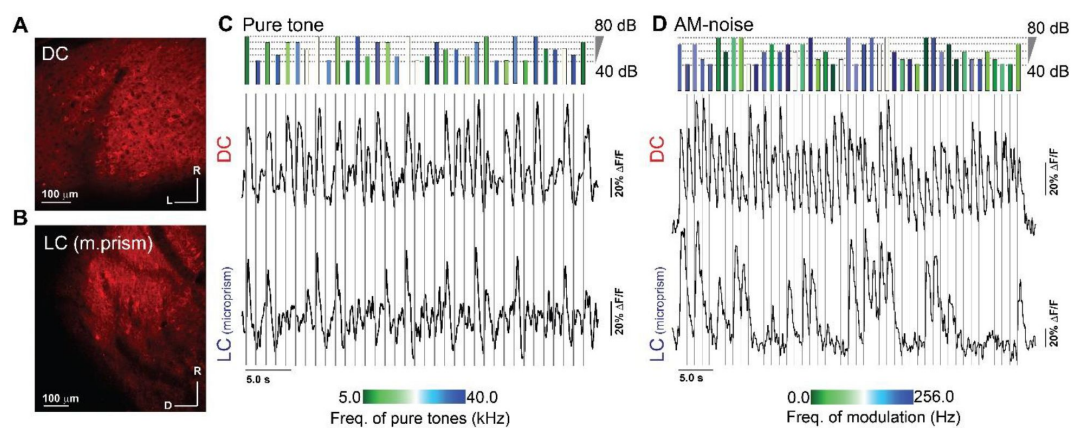


Figure 3

2P expression of jRGECO1a along with evoked signals in the DC imaged from the dorsal surface and LC_(microprism).

(A-B) The 2P images of the jRGECO1a expression on the DC imaged from the dorsal surface and the LC_(microprism), respectively. (C-D) The time traces of the evoked calcium signals by either different frequencies and amplitudes of pure tones or different overall amplitudes and rates of AM of broadband noise, respectively, which were obtained from two different cells on the DC (top traces) or the LC_(microprism) (bottom traces); The length of the colored bars (top) indicates the intensity of the stimulus, while the color indicates the tone or AM frequency depending on the color scale (bottom). AM-noise: Amplitude modulated broadband noise, D: Dorsal, DC: Dorsal cortex, L: Lateral, LC: Lateral cortex, R: Rostral.

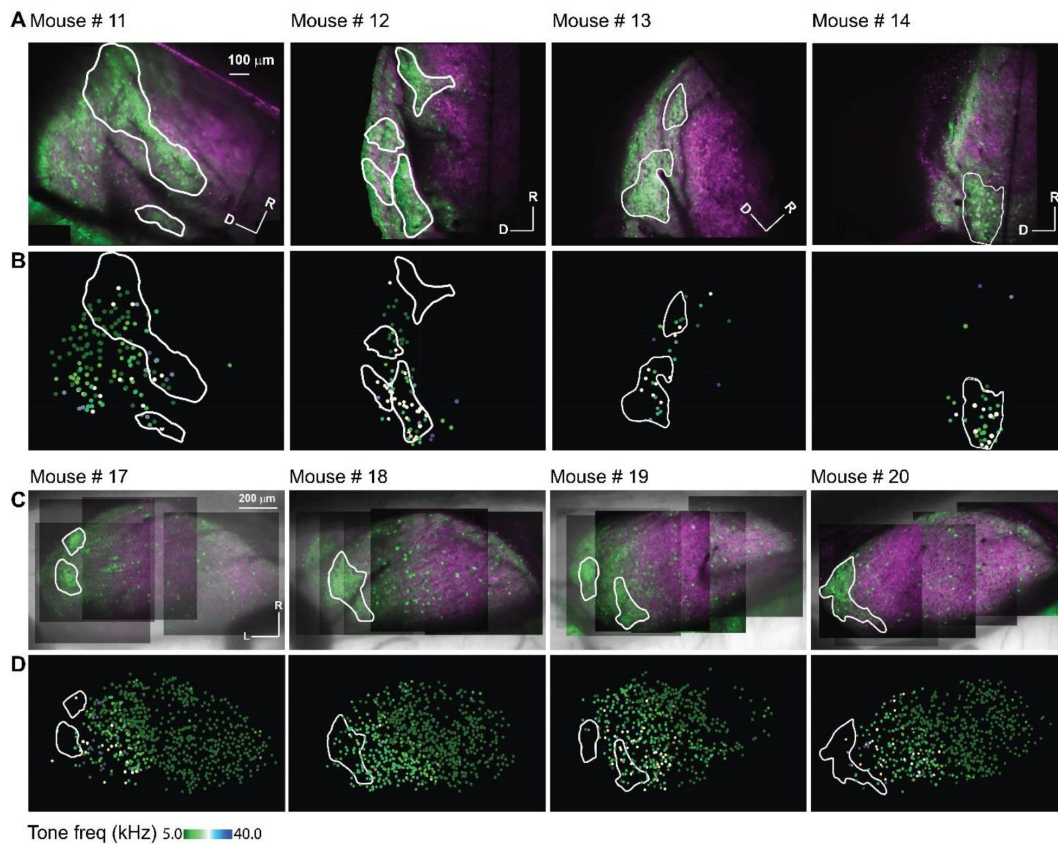


Figure 4

The auditory cellular organization of the LC_(microprism) vs. DC imaged from the dorsal surface.

(A and C) The 2P images of GFP (green) and jRGECO1a expression (red) on either the LC_(microprism) or the DC imaged from the dorsal surface, respectively. (B and D) The pseudocolor images show the responsive cells to the pure tone of different combinations of frequencies and levels on the surface of the LC_(microprism) or the DC imaged from its dorsal surface, respectively. D: Dorsal, L: Lateral, R: Rostral.

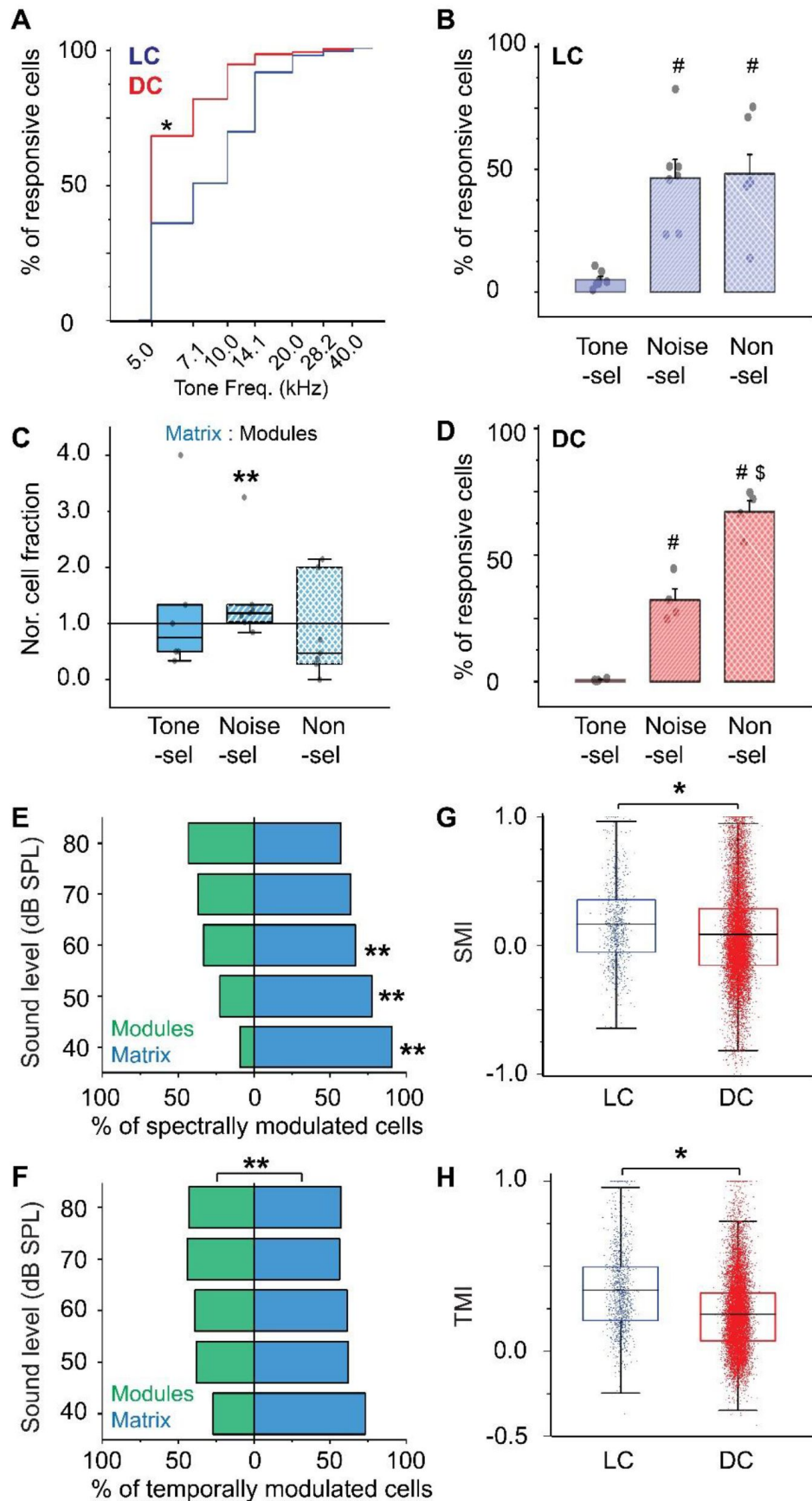


Figure 5

The cellular response to pure tones and AM-noise of the LC_(microprism) vs. DC imaged from the dorsal surface.

(A) The cumulative distribution function of the best tone frequencies of all cells collected either from the LC imaged via microprism (blue line) or the DC imaged directly from its dorsal surface (red line) (Mann-Whitney Test: $z = -11.34$, $p < 0.001$, a median of best-tuned frequency = 5 and 7.1 Hz for the DC (2484 cells from 4 animals) and the LC_(microprism) (445 cells from 7 animals), respectively, $*p < 0.05$, DC vs LC_(microprism)). (B) A bar graph showing the fractions of cells responding to only pure tone (Tone-sel), only AM-noise (Noise-sel), or to both nonselectively (Non-sel) on the LC_(microprism) (7 animals) (one way-ANOVA, $f_{(2,18)} = 14.9$, $p = 1.6 \times 10^{-4}$, Fisher posthoc test: $p = 2.0 \times 10^{-4}$ and 1.3×10^{-4} for Noise-sel and Non-sel vs Tone-sel, respectively, $p = 0.84$ for Noise-sel vs Non-sel, % of responsive cells $\pm$ SEM = $5 \pm 1\%$, $46 \pm 7\%$, and $48 \pm 7\%$ for Tone-sel, Noise-sel, and Non-sel, respectively, $*p < 0.05$ vs Tone-sel). (C) Box graphs showing the fractions of Tone-sel, Noise-sel, and Non-sel cells within modules and matrix of the LC_(microprism). (D) A bar graph showing the fractions of Tone-sel, Noise-sel, or Non-sel cells on the DC imaged directly from the dorsal surface (4 animals) (one way-ANOVA, $f_{(2,9)} = 86.9$, $p = 1.2 \times 10^{-6}$, Fisher posthoc test: $p = 1.5 \times 10^{-4}$ and 3.4×10^{-7} for Noise-sel and Non-sel vs Tone-sel, respectively, $p = 7.0 \times 10^{-5}$ for Noise-sel vs Non-sel, % of responsive cells $\pm$ SEM = $0.6 \pm 0.2\%$, $32 \pm 4\%$, and $67 \pm 4\%$ for Tone-sel, Noise-sel, and Non-sel, respectively, $*p < 0.05$ vs Tone-sel and $*p < 0.05$ vs Noise-sel). (E and F) Bar graphs showing the percentage of spectrally (two-way ANOVA: $f_{(1,4,20)} = 50.3$, 0.0, and 4.0 – $p = 7.1 \times 10^{-7}$, 1.0, and 0.01 for the region, sound level, and interaction, respectively, Fisher post hoc, $p = 5.1 \times 10^{-6}$, 4.9×10^{-4} , 0.02, 0.058, and 0.14 for modules vs. matrix at 40, 50, 60, 70, and 80 dB, respectively, % of spectral modulated cells $\pm$ SEM (modules vs matrix) = 9 ± 1 vs $90 \pm 1\%$ at 40 dB, 22 ± 4 vs $77 \pm 4\%$ at 50 dB, 33 ± 10 vs $66 \pm 10\%$ at 60 dB, 36 ± 13 vs $63 \pm 13\%$ at 70 dB, and 43 ± 11 vs $56 \pm 11\%$ at 80 dB, $n = 3$ animals) or temporally (two-way ANOVA: $f_{(1,4,20)} = 19.3$, 0.0, and 1.2 – $p = 7.1 \times 10^{-7}$, 1.0, and 0.32 for the region, sound level, and interaction, respectively, % of temporally modulated cells $\pm$ SEM = 38 ± 3 and $61 \pm 3\%$ for modules vs matrix, $n = 3$ animals) modulated cells, respectively, across different sound levels in modules (green bars) vs. matrix (blue bars) imaged via microprism, ($**p < 0.05$, matrix vs. modules). (G and H) Box plots showing the mean (black lines) and the distribution (colored dots) of the SMI (two-way ANOVA: $f_{(1,4,14583)} = 48.4$, 210.4, and 21.3 – $p < 0.001$ for the region, sound level, and interaction, respectively, Fisher post hoc test, $p = 6.2 \times 10^{-4}$, 4.2×10^{-5} , < 0.001 , and < 0.001 for LC vs. DC at 40, 60, 70, and 80 dB respectively, mean of SMI $\pm$ SEM: 0.16 ± 0.01 and 0.08 ± 0.003 for LC vs DC, $n = 1020$ cells from 7 animals (LC_(microprism)), and 13574 cells from 4 animals (DC)) and TMI (two way ANOVA: $f_{(1,4,16139)} = 431.4$, 59.1, and 8.3 – $p < 0.001$ for the region, sound level, and interaction, respectively, Fisher post hoc test, $p = 0.002$, < 0.001 , < 0.001 , < 0.001 , and < 0.001 for LC vs. DC at 40, 50, 60, 70, and 80 dB, respectively, mean of TMI $\pm$ SEM: 0.35 ± 0.006 and 0.21 ± 0.001 for LC vs DC, $n = 1441$ cells from 7 animals (LC_(microprism)), and 14708 cells from 4 animals (DC)), respectively, across different sound levels in the LC_(microprism) (blue) vs. the DC (red), $*p < 0.05$, LC vs. DC). DC: Dorsal cortex, LC: Lateral cortex. SMI: spectral modulation index, TMI: temporal modulation index.

Detection of AM is critical to identify natural sounds as well as to aggregate sound sources during auditory scene analysis, which is important for communication (Joris, Schreiner, & Rees, 2004; Rees & Langner, 2005; Singh & Theunissen, 2003). The IC also is the first site in the ascending auditory pathway where a substantial number of neurons are rate-tuned to AM (Frisina, 2001; Joris et al., 2004; Langner, 1992; Nelson & Carney, 2007; Rabang et al., 2012). Even though calcium indicators do not provide a spike-by-spike readout of neural activity, their integrated signal does correlate with the number of spikes produced by a neuron over short time windows (Vogelstein et al., 2010; Yaksi & Friedrich, 2006). Thus, this integrated signal can be used to measure rate responses to AM. Using this integrated signal, we tested the ability of the cells of the LC_(microprism) to respond to broadband noise at different AM rates within each cellular motif. Within the fraction of responsive cells, the LC_(microprism) showed cells of different response profiles. While some cells responded only to pure tones or tone-selective (Tone-sel), other cells responded only to AM-noise or noise-selective (Noise-sel). However, some cells responded to both pure tones and AM-noise (Non-sel), when the two stimuli were individually presented. While the

LC_(microprism) had similar fractions of Noise-sel and Non-sel cells, both fractions were significantly higher than that of the Tone-sel cells (**Figure 5B** [↗](#), one-way-ANOVA, $p = 1.6 \times 10^{-4}$). We also observed that modules and matrix differentially process the spectral information. For instance, the matrix had a significantly greater fraction of Noise-sel cells compared to the modules (**Figure 5C** [↗](#), Mann-Whitney Test, $z = -2.32$, $p = 0.02$). In contrast, both cellular motifs had similar fractions of Tone-sel and Non-sel cells (**Figure 5C** [↗](#), Mann-Whitney Test: $z = 0.48$, $p = 0.62$ for Tone-sel and $z = 1.36$, $p = 0.17$ for Non-sel). These findings suggest that neurons in the matrix are more responsive to sounds of greater spectral and temporal complexity than those in the modules. Similar to the LC, the DC showed significantly higher fractions of Noise-sel and Non-sel cells than Tone-sel cells, but the cellular fraction of Non-sel cells was higher than that of Noise-sel cells (**Figure 5D** [↗](#), one way-ANOVA, $f_{(2,9)} = 86.9$, $p = 1.2 \times 10^{-6}$).

By pooling the cells based on their response to either pure tones or AM-noise, animals showed a lower fraction of responsive neurons to the acoustic stimulation in the LC_(microprism) compared to the DC regardless of the sound type (two-way ANOVA, $f_{(1,1,18)} = 9.1$, 7.8, and 0.1, $p = 0.007$, 0.01, and 0.75 for region (LC vs DC), stimulus (pure tones vs AM-noise), and interaction, respectively, $n = 7$ animals for LC and 4 animals for DC, the % of responsive cells $\pm$ SEM = $48 \pm 7\%$ for LC and $75 \pm 2\%$ for DC). To evaluate if any optical artifacts due to imaging through a prism could be responsible for this finding, the number of responsive cells imaged from the lateral-most dorsal surface of the IC, which we consider part of the LC because of the presence of GABAergic modules (LC imaged from the dorsal surface or LC_(dorsal surface)), was quantified and compared with that obtained from the LC_(microprism). We observed that there were no significant differences between the number of tone or noise-responsive neurons between the LC_(microprism) or LC_(dorsal surface) (two-way ANOVA, $f_{(1,1,18)} = 0.07$, 14.1, and 2.11, $p = 0.79$, 0.001, and 0.16 for the region, stimulus (pure tones vs AM-noise), and interaction, respectively, $n = 7$ animals for LC_(microprism) and 4 animals for LC_(dorsal surface)), the % of responsive cells $\pm$ SEM = $48 \pm 7\%$ for LC_(microprism) and $45 \pm 11\%$ for LC_(dorsal surface)).

LC and DC respond differently to spectral and temporal modulation of acoustic inputs

Given that more cells from LC_(microprism) and DC had responses to AM-noise compared to pure tones, we further quantified the selectivity to spectral or temporal modulation by computing the spectral modulation index (SMI) or the temporal modulation index (TMI), respectively, as well as the number of modulated cells for all the responsive cells (see Methods) across the tested sound levels.

For the LC_(microprism), the matrix has a greater proportion of spectrally modulated cells (cells responding only to pure tones and unmodulated noise) than modules. The higher fractions of modulated cells in matrix were most prominently at sound levels below 70 dB (**Figure 5E** [↗](#), two-way ANOVA: $p = 7.1 \times 10^{-7}$, 1.0, and 0.01 for the region, sound level, and interaction, respectively) suggesting that the cells in the matrix have a lower threshold to detect the spectral features of sound. Similarly, the matrix had more cells responsive to temporal modulation (cells responding to AM-noise) as a main effect (**Figure 5F** [↗](#), two-way ANOVA, $p = 2.7 \times 10^{-4}$, 1.0, and 0.32 for the region, sound level, and interaction, respectively), which was consistent with the above results (**Figure 5C** [↗](#)).

The DC and LC imaged via microprism were found to have different profiles of modulation of responses to spectral and temporal complexity. Although there is a general trend for SMI and TMI to decrease with increasing sound-pressure level, corresponding to a decrease in the response to pure-tone and AM-noise, respectively, overall there is a statistically-significant higher level of average SMI and TMI in the LC compared to the DC (**Figures 5G and H** [↗](#), SMI: two-way ANOVA, p

< 0.001 for all region, sound level, and interaction, TMI: two-way ANOVA, $p < 0.001$ for all regions, sound level, and interactions). These results suggested that neurons in the LC are more sensitive to spectral and temporal modulation than those in the DC.

LC modules and matrix differentially respond to acoustic and somatosensory stimuli

Somatosensory inputs from many cortical and subcortical structures have been identified in the non-lemniscal division of the IC, particularly in the LC (A. M. Lesicko et al., 2016; Lohse, Zimmer-Harwood, Dahmen, & King, 2022; Zhou & Shore, 2006). Previous work has shown that while somatosensory projections mostly target the modules, the auditory projections tend to target the matrix (A. M. Lesicko et al., 2016), which indicates that the LC is a multisensory integration site. Therefore, we examined the somatosensory and auditory cellular responses on the surface of the LC by coupling the 2P imaging with the microprism with a whisker or acoustic stimulation (see Methods). Acoustic stimulation was done with 500 ms of unmodulated broadband noise at 80 dB SPL. After the acoustic or somatosensory stimulation, evoked cellular responses were observed from the neurons of the LC_(microprism) as well as the DC imaged directly from the dorsal surface. According to these cellular responses, different types of neurons were found in modules and matrix of the LC_(microprism) as well as the DC (**Figure 6A-D**). Also, the auditory response evoked by auditory stimulation only was found to be modulated (suppressed vs enhanced) when the auditory stimulation was simultaneously presented with somatosensory stimulation (bimodal stimulation) (**Figure 6E and F**). The cells of every structure were then mapped according to their responses. Auditory and somatosensory responses were evoked in the neurons of both matrix and modules imaged via microprism (**Figures 7A**, B, and D). Within the pool of responsive cells, the cells were categorized into cells that respond only to auditory stimulation [auditory selective cells (Aud-sel)], only to somatosensory stimulation [somatosensory selective cells (Som-sel)], or nonselectively to both stimuli (Aud/Som-nonsel) (**Figure 7F**). Both modules and matrix showed a higher fraction of Aud-sel cells than that of Som-sel and Aud/Som-nonsel cells (**Figure 7H**, Chi-Square test, $\chi^2 = 33.1$, $p = 3.5 \times 10^{-6}$). Pooling the cells based on their auditory vs somatosensory responses revealed that more auditory-responsive cells (Aud-sel + Aud/Som-nonsel) were found in the matrix, while more somatosensory-responsive cells (Som-sel + Aud/Som-nonsel) were clustered in the modules (**Figures 7B**, D, F, and I, Chi-Square test, $\chi^2 = 26.0$, $p = 9.3 \times 10^{-5}$), which was consistent with the reported difference in inputs to modules compared to matrix (A. M. Lesicko et al., 2016). Additionally, while most of the acoustically-responsive cells in the matrix showed onset responses, the cells showing acoustically-driven offset responses were clustered in the modules (**Figures 7C and J**, Chi-Square test, $\chi^2 = 120.5$, $p < 0.001$). Moreover, a higher fraction of Aud/Som-nonsel cells in the modules had offset, not onset, auditory response than those located in the matrix (Chi-Square test, $\chi^2 = 13.2$, $p = 0.004$, % of Aud/Som-nonsel cells = 57%, 15%, 8%, and 20%, for auditory offset cells in modules, auditory onset cells in modules, auditory offset cells in matrix, and auditory onset cells in the matrix, respectively), which suggests a multisensory integrative function by those cells as discussed later. Given the specific spatial location of the somatosensory-responsive cells in the modules which contain clusters of GABAergic cells, we asked if somatosensory stimulation could suppress the auditory response when both stimuli are presented simultaneously (bimodal stimulation). To test this hypothesis, we calculated and visualized the response index (RI), which was calculated as the ratio of the cellular response to bimodal stimulation to the cellular response to acoustic response alone for auditory-responsive cells. Accordingly, some cells showed suppressed (RI < 1) or enhanced (RI > 1) auditory responses in both modules and matrix following bimodal stimulation. We observed that matrix and modules have more cells with suppressed auditory responses than those having enhanced auditory responses. However, matrix had a higher fraction of cells with suppressed responses than modules, while modules had a higher fraction of cells with enhanced responses (**Figures 7G and K**, Chi-Square test, $\chi^2 = 12.9$, $p = 0.04$).

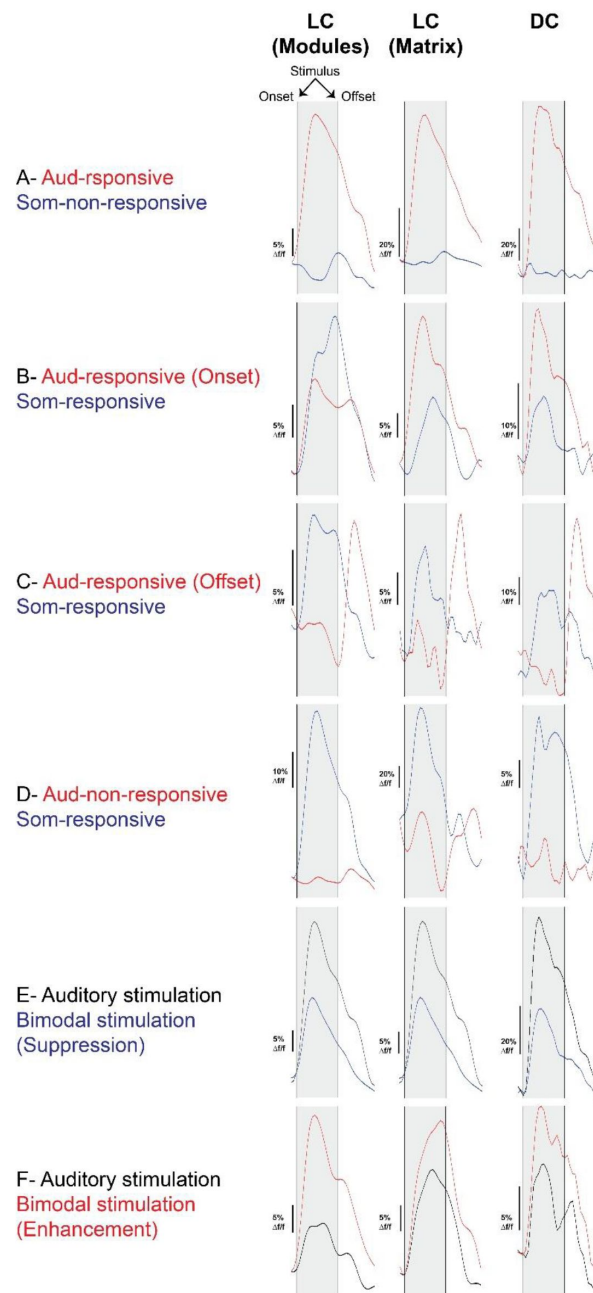


Figure 6

The waveforms of the somatosensory vs. acoustic responses of the LC_(microprism) and the DC. imaged via microprism.

Examples of different cell types based on their responses to auditory or somatosensory stimulations as well as their response timing to the auditory stimulation as indicated by waveforms of the evoked calcium signals.

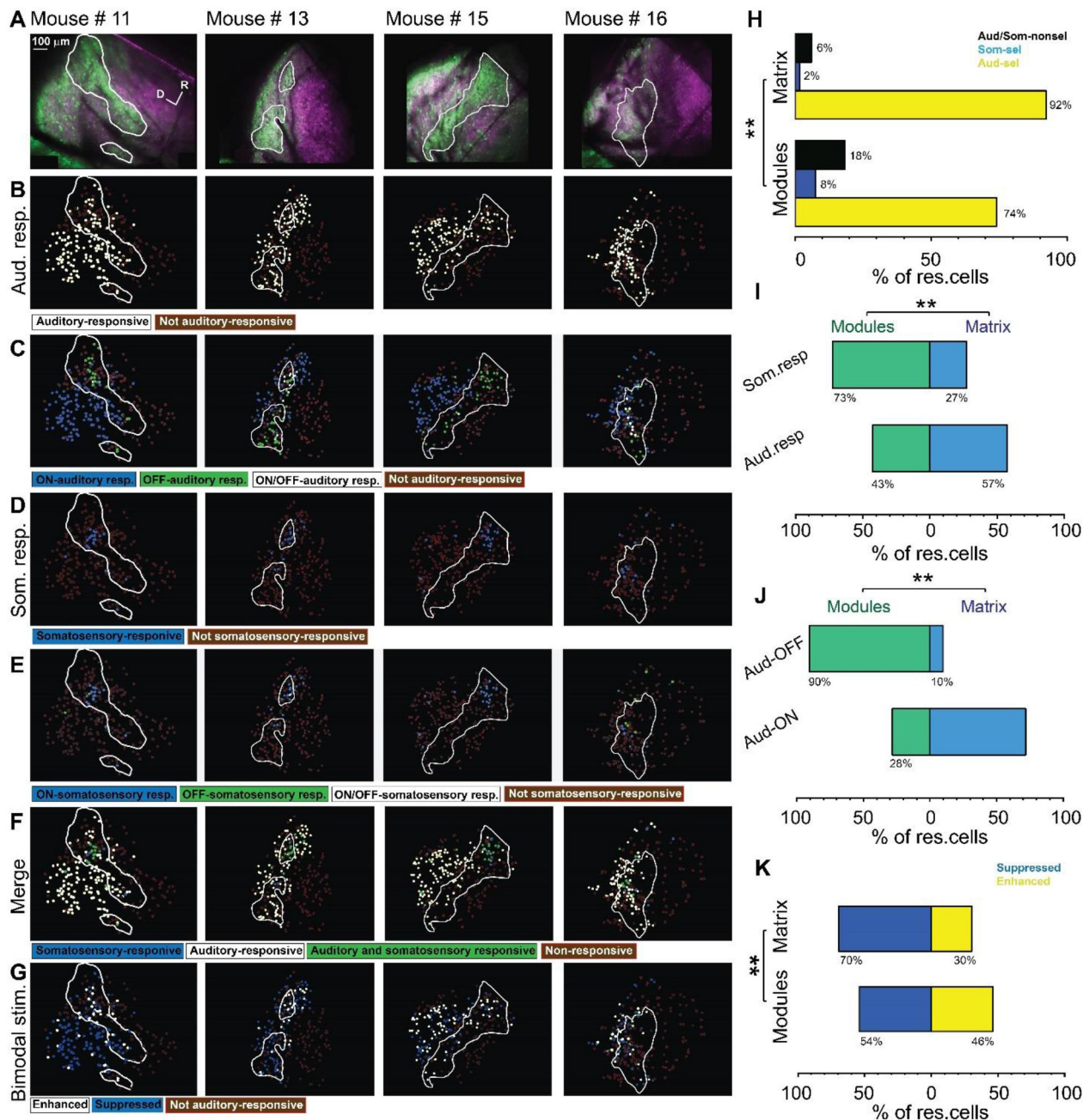


Figure 7

Types of somatosensory and auditory responses from the LC neurons imaged via microprism.

(A) The 2P images of GFP (green) and jRGECO1a expression (magenta) in the LC_(microprism) show the modules within white irregular lines. (B, D, and F) The pseudocolor images show the auditory (white circles) or somatosensory (blue circles) responsive cells of the LC_(microprism) as well as the merged response (green circles), respectively. (C and E) The pseudocolor images show onset (ON) (blue circles), offset (OFF) (green circles), or onset/offset (ON/OFF) (white circles) responsive cells of the LC_(microprism) to auditory and somatosensory stimuli, respectively. (G) The pseudocolor images show enhanced (white circles) and suppressive response (blue circles) of the auditory responsive cells of the LC_(microprism) following the bimodal stimulation based on their response index (RI). (H) A bar graph showing the percentage of responsive cells to auditory stimulation only or auditory selective cells (Aud-Sel, yellow bars), to somatosensory stimulation only or somatosensory selective cells (Som-sel, blue bars), or nonselectively to both auditory and somatosensory stimulations (Aud/Som-nonsel, black bars) in modules vs matrix (Chi-Square test, $\chi^2 = 33.1$, $p = 3.5 \times 10^{-6}$, % of responsive cells = 74%, 8%, and 18% for Aud-sel, Som-sel, and Aud/Som-nonsel, respectively at (Modules, 235 cells) vs 92%, 2%, and 6% for Aud-sel, Som-sel, and Aud/Som-sel, respectively at (matrix, 297 cells) – 4 animals, $**p < 0.05$ vs modules). (I) A bar graph showing the percentage of all cells having auditory responses (Aud. resp) or those having somatosensory responses (Som. resp) in modules vs matrix (Chi-Square test, $\chi^2 = 26.0$, $p = 9.3 \times 10^{-5}$, % of responsive cells (Modules vs matrix) = 43% vs 57% for (Aud-resp, 509 cells) and 73% vs 27% for (Som. resp, 84 cells) – 4 animals, $**p < 0.05$ vs modules). (J) A bar graph showing the percentage of auditory responsive cells with onset (Aud-ON) vs those with offset (Aud-OFF) responses in modules vs matrix (Chi-Square test, $\chi^2 = 120.5$, $p < 0.001$, % of responsive cells (Modules vs matrix) = 28% vs 72% for (Aud-ON, 390 cells) and 90% vs 10% for (Aud-OFF, 93 cells) – 4 animals, $**p < 0.05$ vs modules). (K) A bar graph showing the percentage of auditory responsive cells with suppressed vs those with enhanced responses following bimodal stimulation in modules vs matrix (Chi-Square test, $\chi^2 = 12.9$, $p = 0.04$, % of responsive cells = 54% vs 46% for suppressed vs enhanced (modules, 217 cells) and 70% vs 30% for suppressed vs enhanced (matrix, 292 cells) – 4 animals, $**p < 0.05$ vs modules). D: Dorsal, R: Rostral.

Consistent with the above data, the LC_(dorsal surface) (Figures 8A-D, dorsolateral white rectangle) shared common response profiles with the LC_(microprism). For instance, both regions had a higher percentage of somatosensory responsive cells than the DC (Figures 8A-D, dorsal yellow and dorsomedial blue rectangles) as quantitatively shown (Figure 8H, one-way ANOVA, $p = 0.02$), which suggests that the LC is the main region of the IC shell that processes somatosensory information, consistent with the anatomical reports (Aitkin, Dickhaus, Schult, & Zimmermann, 1978; Jain & Shore, 2006; A. M. Lesicko et al., 2016; Liu et al., 2023). Similar to the LC_(microprism), the matrix and modules imaged from the dorsal surface had more percentage of Aud-sel cells than Som-sel and Aud/Som-nonsel cells (Figures 8E and I, Chi-Square test, $\chi^2 = 86.1$, $p < 0.001$). Collectively, the matrix had more auditory responsive cells than modules, while modules had more somatosensory responsive cells than matrix (Figure 8J, Chi-Square test, $\chi^2 = 66.5$, $p < 0.001$). Additionally, more cells with onset auditory response were found in the matrix, while the cells with offset auditory responses were clustered in the modules (Figures 8F and K, Chi-Square test, $\chi^2 = 500.2$, $p < 0.001$).

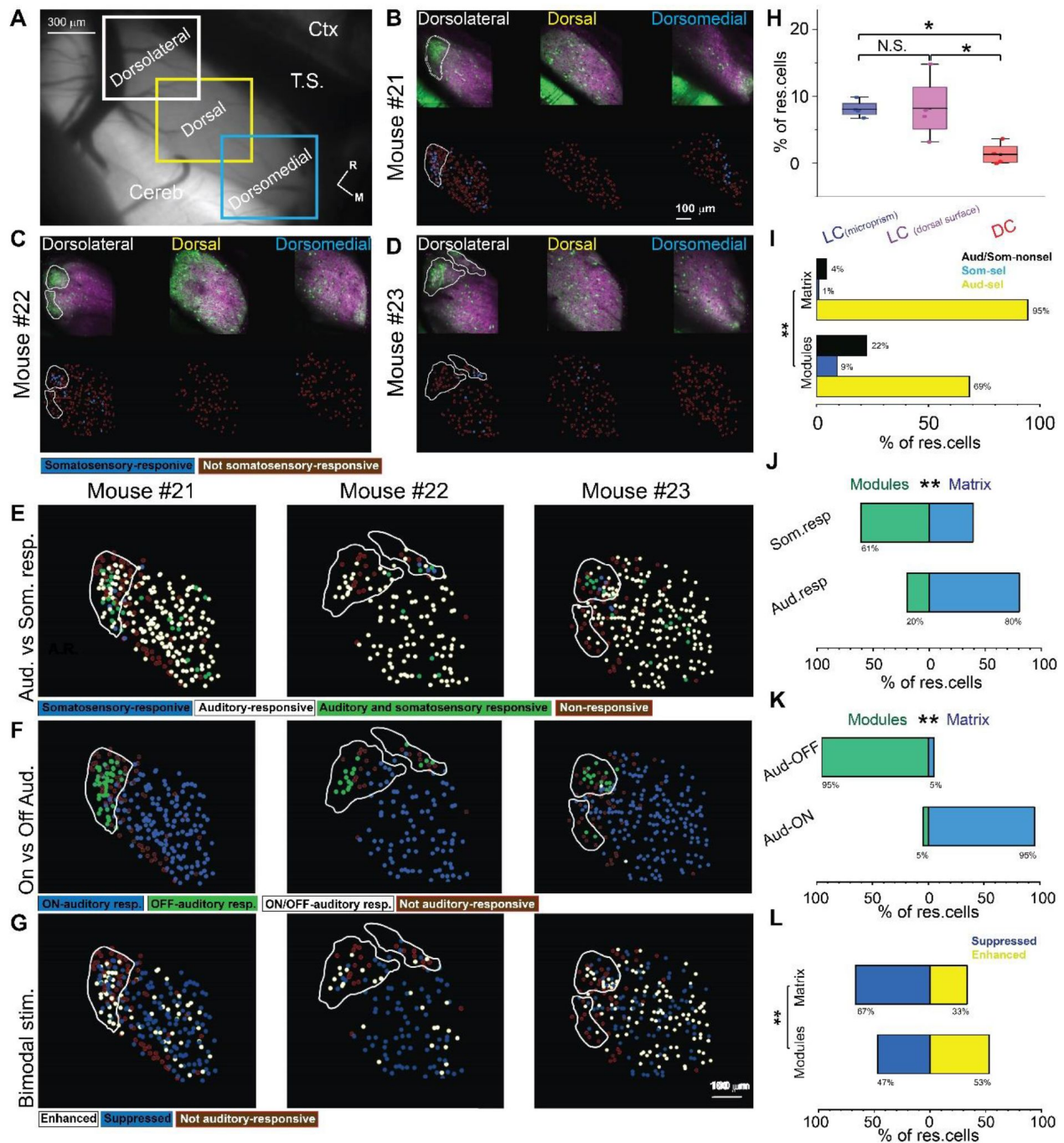


Figure 8

The somatosensory vs. auditory responses over the DC neurons.

(A) A low magnification of the image obtained from the IC surface. (B-D) The top row: The 2P images of GFP (green) and jRGECO1a expression (magenta) of three main regions (dorsolateral, dorsal, and dorsomedial) of the DC surface imaged from the dorsal surface showing the modules within irregular white lines in the dorsolateral region, the bottom row: The pseudocolor images show the somatosensory responsive cells (blue circles) after somatosensory stimulations. (E) The pseudocolor images show the responsive cells of the LC_(dorsal surface) to auditory (white circles), somatosensory (blue circles), or both stimulations (green circles). (F) The pseudocolor images show auditory onset (Aud-ON – blue circles), offset (Aud-OFF, green circles), or onset/offset (Aud-ON/OFF, white circles) responsive cells of the LC imaged from the dorsal surface to auditory stimulation. (G) The pseudocolor images show the enhanced (white circles) and suppressive (blue circles) response of the auditory responsive cells of the LC imaged from the dorsal surface following the bimodal stimulation based on their response index (RI). (H) A box plot shows the percentage of somatosensory responsive cells in the LC_(microprism), the LC_(dorsal surface) or the DC (one-way ANOVA, $p = 0.02$, the percentage of somatosensory responsive cells mean $\pm$ SEM = $1.3 \pm 0.83\%$ (DC, 4 animals), $8.2 \pm 2.4\%$ (LC_(dorsal surface), 4 animals), and $8.1 \pm 0.64\%$ (LC_(microprism), 4 animals, $*p < 0.05$, vs. DC). (I) A bar graph showing the percentage of responsive cells to auditory stimulation only or auditory selective cells (Aud-Sel, yellow bars), to somatosensory stimulation only or somatosensory selective cells (Som-sel, blue bars), or nonselectively to both auditory and somatosensory stimulations (Aud/Som-nonsel, black bars) in modules vs matrix (Chi-Square test, $\chi^2 = 86.1$, $p < 0.001$, % of responsive cells = 69%, 9%, and 22% for Aud-sel, Som-sel, and Aud/Som-nonsel (modules, 152 cells) and 95%, 1%, and 4% % for Aud-sel, Som-sel, and Aud/Som-nonsel (matrix, 573 cell) - 4 animals, $**p < 0.05$ vs modules). (J) A bar graph showing the percentage of all cells having auditory responses (Aud. resp) or those having somatosensory responses (Som. resp) in modules vs matrix (Chi-Square test, $\chi^2 = 66.5$, $p < 0.001$, % of responsive cells (Modules vs matrix) = 20% vs 80% for (Aud. resp, 705 cells) and 61% vs 39% for (Som. resp, 79 cells) - 4 animals, $**p < 0.05$ vs modules). (K) A bar graph showing the percentage of auditory responsive cells with onset (Aud-ON) vs those with offset (Aud-OFF) responses in modules vs matrix (Chi-Square test, $\chi^2 = 500.2$, $p < 0.001$, % of responsive cells (Modules vs matrix) = 5% vs 95% for (Aud-ON, 584 cells) and 95% vs 5% for (Aud-OFF, 116 cells) - 4 animals, $**p < 0.05$ vs modules). (L) A bar graph showing the percentage of auditory responsive cells with suppressed vs those with enhanced responses following bimodal stimulation in modules vs matrix (Chi-Square test, $\chi^2 = 18.5$, $p = 1.6 \times 10^{-5}$, % of responsive cells = 47% vs 53% for suppressed vs enhanced (modules, 138 cells) and 67% vs 33% for suppressed vs enhanced (matrix, 567 cells) - 4 animals, $**p < 0.05$ vs modules). Cereb: Cerebellum, Ctx: Cerebral cortex, M: medial, R: Rostral, TS: Transverse sinus.

Consistent with the LC_(microprism), some cells had suppressed or enhanced auditory responses following bimodal stimulation (**Figure 8** [figure supplement 1](#)). The matrix imaged from the dorsal surface showed also more cells with suppressed auditory responses than those having enhancing auditory responses induced by bimodal stimulation. Additionally, the matrix had a higher fraction of cells with suppressed responses than modules, while modules had a higher fraction of cells with enhanced responses than matrix. In contrast to the LC_(microprism), the modules imaged directly from the dorsal surface had a higher fraction of cells with enhanced responses than those with suppressed responses, which may indicate that the ratio between suppressed and enhanced responses in modules could be altered based on the dorsoventral location (**Figures 8G and L** [figure supplement 1](#), Chi-Square test, $\chi^2 = 18.5$, $p = 1.6 \times 10^{-5}$). In general, these data confirmed the findings obtained from the LC imaged via microprism and emphasized the functional difference between the LC and DC.

LC and DC differ in their acoustic and multimodal response properties in awake animals

To determine if the presence of ketamine/xylazine anesthesia influenced the response properties detailed above, additional experiments were performed in awake animals ($n = 6$ total, 3 for the LC_(micropism) and 3 for the DC). Similar to what was observed under anesthesia, the cells of the LC_(micropism) did not show clear evidence of tonotopy (**Figure 9B**), which was indicated by a low R^2 of the regressions fit between the BTFs of the cells and their locations over different anatomical axes (**Figure 9**-figure supplement 1A). Similarly, the cells of the modules were tuned to relatively higher frequencies than those in matrix (Mann-Whitney Test, $z = 3.8$, $p = 1.0 \times 10^{-4}$, median of best frequency = 10 vs 7.1 kHz for modules (148 cells from 3 animals) vs matrix (287 cells from 3 animals)). Additionally, the data obtained from imaging the DC under awake preparations showed a similar tonotopic map (**Figure 9D**) as that observed under anesthesia indicated by a high R^2 of the regression fit between the BTFs of the cells and their locations over the rostromedial and caudolateral axis (**Figure 9**-figure supplement 1B). Yet, a greater proportion of high-frequency cells were visually observed in the mediodorsal portion of the DC, which was indicated by a higher R^2 of the nonlinear quadratic fit than that of the linear fit, leading to the observation of a narrow low-frequency zone similar to the previous work under awake preparations (A. B. Wong & Borst, 2019). The LC_(micropism) had also more cells tuned to higher frequencies compared to the DC (**Figure 10A**, Mann-Whitney test, $p = 1.6 \times 10^{-6}$). We also examined the width of the receptive field (RF) of the DC vs LC neurons at each tested sound level under anesthetized and awake preparations using the binarized receptive field sum (RFS) method (Bowen, Winkowski, & Kanold, 2020). Both DC and LC of the anesthetized animals had higher RFS mean and median than those of awake animals given that ketamine was known to broaden the RF (Guo et al., 2012). In both preparations (anesthetized and awake), the DC had a higher RFS mean than that of the LC (**Figure 10**-figure supplement 1, Kruskal-Wallis ANOVA, $p < 0.001$), which could be consistent with the finding that the DC had a lower SMI than LC.

Similar to what was shown in the anesthetized animals, the fraction of Noise-sel cells was higher than that of Tone-sel cells in both LC_(micropism) and DC imaged directly from the top in awake animals (**Figure 10B and D**). In contrast to the anesthetized animals, the fraction of Non-sel cells had different profiles. For instance, while the percentage of Non-sel cells was higher than Noise-sel and Tone-sel cells in the LC_(micropism) (**Figure 10B**, one way ANOVA, $f_{(2,6)} = 41.6$, $p = 3.0 \times 10^{-4}$), they were significantly lower than that of Noise-sel cells and similar to that of Tone-sel cells in the DC (**Figure 10D**, one way ANOVA, $f_{(2,6)} = 12.8$, $p = 0.006$). Despite that modules and matrix had similar fractions of Noise-sel, Tone-sel, and Non-sel cells (**Figure 10C**, Mann-Whitney Test (Modules vs matrix), $z = -0.46$, -0.46 , and -1.85 , $p = 0.64$, 0.64 , and 0.06 for Noise-sel, Tone-sel, and Non-sel, respectively), matrix had more spectrally (**Figure 10E**, two-way ANOVA: $p = 5.3 \times 10^{-10}$, 1.0 , and 0.17 for the region, sound level, and interaction, respectively) and temporally (**Figure 10F**, two-way ANOVA: $p = 4.6 \times 10^{-7}$, 1.0 , and 0.11 for the region, sound level, and interaction, respectively) modulated cells than modules, which was similar to anesthetized animals. Additionally, the LC_(micropism) and DC showed different profiles than those found by anesthetized animals. While LC and DC had similar SMI (**Figure 10G**, two-way ANOVA: $p = 0.22$, 2.1×10^{-12} , and 1.1×10^{-6} for the region, sound level, and interaction, respectively), the LC_(micropism) had a lower average of TMIs compared to the DC (**Figure 10H**, two-way ANOVA: $p = 1.9 \times 10^{-0}$, 6.6×10^{-24} , and 0.01 for the region, sound level, and interaction, respectively).

Similar to the anesthetized preparations, the auditory and somatosensory responsive cells were topographically organized across modules and matrix (**Figures 11A**, C, and E). However, it was noticed that the somatosensory and auditory responsive cells were clustered within a pool of nonresponsive cells (**Figure 11E**). Some of these nonresponsive cells to unmodulated broadband noise (**Figure 11**-supplementary figures 1A) were found to respond to pure tones (**Figure 11**-supplementary figures 1B, blue arrows) indicating that these nonresponsive cells may be sensitive

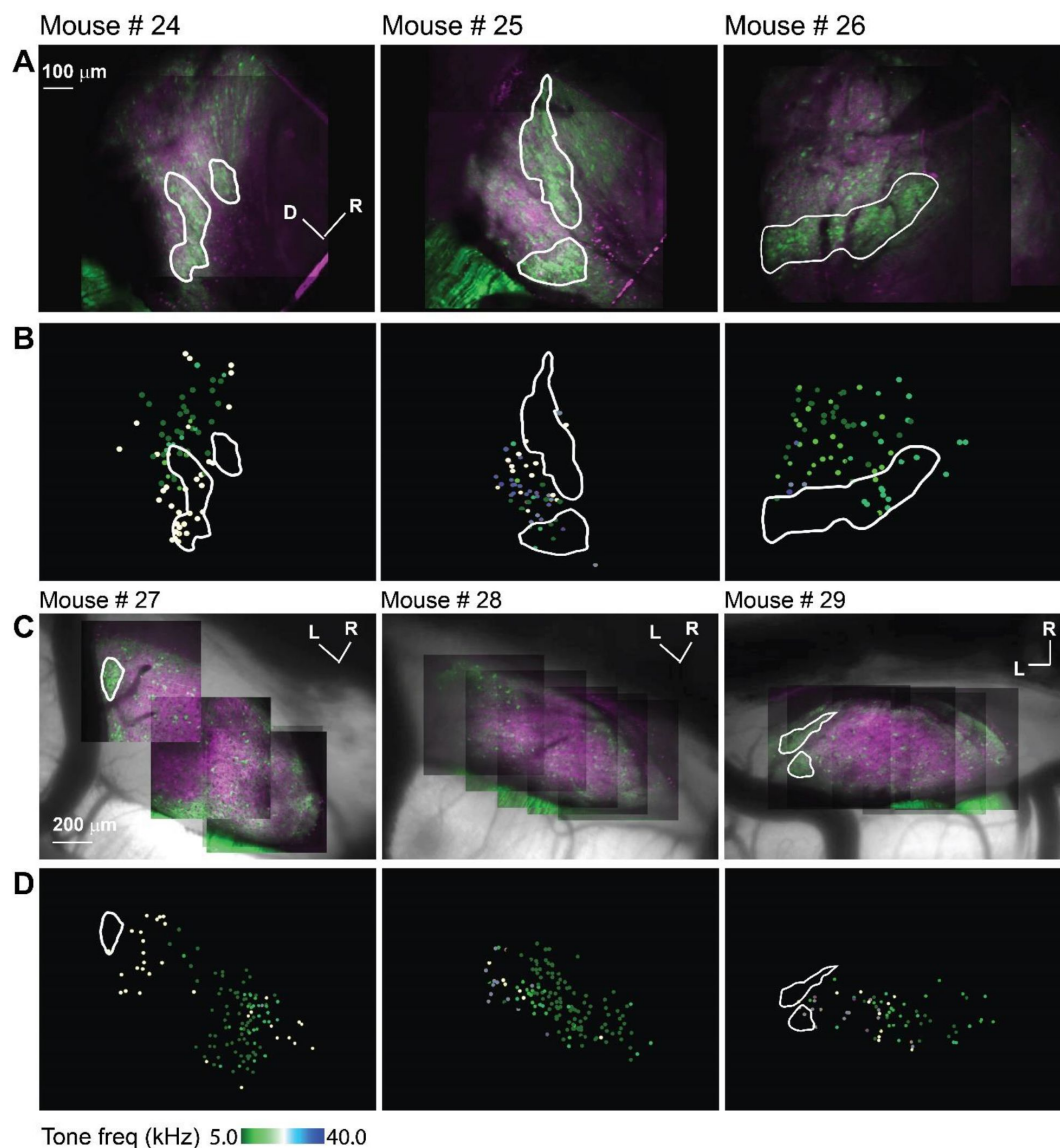


Figure 9

The acoustic response from LC_(microprism) and DC imaged from the dorsal surface in awake animals.

(A and C) The 2P images of GFP (green) and jRGECO1a expression (magenta) on either the LC_(microprism) surface imaged via the microprism or the DC imaged directly from the dorsal surface, respectively showing the modules within irregular white lines. (B and D) The pseudocolor images show the responsive cells to the pure tone of different combinations of frequencies and levels of the LC_(microprism) or the DC imaged directly from the dorsal surface, respectively. D: Dorsal, L: Lateral, R: Rostral.

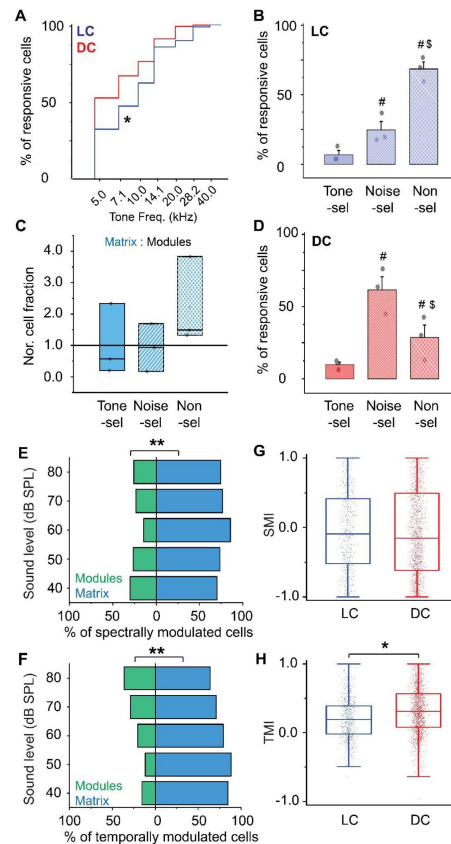


Figure 10

The cellular response to pure tones and AM-noise of the LC_(microprism) vs. DC imaged from the dorsal surface in awake animals.

(A) The cumulative distribution function of the best tone frequencies of all cells collected either from the LC imaged via microprism (blue line) or the DC imaged directly from its dorsal surface (red line) (Mann-Whitney test, $z = -4.8$, $p = 1.6 \times 10^{-6}$, a median of best-tuned frequency = 5 and 10 kHz for the DC (767 cells from 3 animals) and the LC_(microprism) (371 cells from 3 animals), respectively, $*p < 0.05$ vs DC) (B) A bar graph showing the fractions of cells responding to only pure tone (Tone-sel), only AM-noise (Noise-sel), or to both nonselectively (Non-sel) on the LC_(microprism) (3 animals) (one way ANOVA, $f_{(2,6)} = 41.6$, $p = 3.0 \times 10^{-4}$, Fisher posthoc test: $p = 0.04$ and 1.1×10^{-4} for Noise-sel and Non-sel vs Tone-sel, respectively, $p = 7.3 \times 10^{-4}$ for Noise-sel vs Non-sel, % of responsive cells $\pm$ SEM = $6 \pm 3\%$, $24 \pm 6\%$, and 68 ± 5 for Tone-sel, Noise-sel, and Non-sel, respectively, $\#p < 0.05$ vs Tone-sel and $\$p < 0.05$ vs Noise-sel). (C) Box graphs showing the fractions of Tone-sel, Noise-sel, and Non-sel cells within modules and matrix of the LC_(microprism): (D) A bar graph showing the fractions of Tone-sel, Noise-sel, or Non-sel cells on the DC imaged directly from the dorsal surface (3 animals) (one way ANOVA, $f_{(2,6)} = 12.8$, $p = 0.006$, Fisher posthoc test: $p = 0.002$ and 0.12 for Noise-sel and Non-sel vs Tone-sel, respectively, $p = 0.01$ for Noise-sel vs Non-sel, % of responsive cells $\pm$ SEM = $9 \pm 1\%$, $61 \pm 9\%$, and 28 ± 8 for Tone-sel, Noise-sel, and Non-sel, respectively, $\#p < 0.05$ vs Tone-sel and $\$p < 0.05$ vs Noise-sel). (E and F) Bar graphs showing the percentage of spectrally (two-way ANOVA: $f_{(1,4,16)} = 173.0$, 0 , and 1.8 - $p = 5.3 \times 10^{-10}$, 1.0 , and 0.17 for the region, sound level, and interaction, respectively, % of spectrally modulated cells $\pm$ SEM = $23 \pm 2\%$ vs $76 \pm 2\%$ for modules vs matrix - 3 animals, $**p < 0.05$ vs modules) or temporally (two-way ANOVA: $f_{(1,4,16)} = 65.9$, 0 , and 2.2 - $p = 4.6 \times 10^{-7}$, 1.0 , and 0.11 for the region, sound level, and interaction, respectively, % of temporally modulated cells $\pm$ SEM = $23 \pm 2\%$ vs $76 \pm 2\%$ for modules vs matrix, $n = 3$ animals, $**p < 0.05$ vs modules) modulated cells, respectively, across different sound levels in modules (green bars) vs. matrix (blue bars) imaged via microprism, ($**p < 0.05$, matrix vs. modules). (G and H) Box plots showing the mean (black lines) and the distribution (colored dots) of the SMI (two-way ANOVA: $f_{(1,4,2411)} = 1.4$, 15.3 , and 8.3 - $p = 0.22$, 2.1×10^{-12} , and 1.1×10^{-6} for the region, sound level, and interaction, respectively, $SMI \pm SEM = -0.04 \pm 0.02$ vs -0.06 ± 0.01 for LC_(microprism) vs DC - $n = 619$ cells from 3 animals (LC_(microprism)), and 1826 cells from 3 animals (DC)) and TMI (two-way ANOVA: $f_{(1,4,2869)} = 36.2$, 29.3 , and 3.0 - $p = 1.9 \times 10^{-9}$, 6.6×10^{-24} , and 0.01 for the region, sound level, and interaction, respectively, $SMI \pm SEM = 0.2 \pm 0.01$ vs 0.32 ± 0.007 for LC_(microprism) vs DC - $n = 781$ cells from 3 animals (LC_(microprism)), and 2098 cells from 3 animals (DC), $*p < 0.05$ vs DC). DC: Dorsal cortex, LC: Lateral cortex. SMI: spectral modulation index, TMI: temporal modulation index.

to a specific feature or type of sensory stimulations. Additionally, some of the cells that did not respond to any of the tested stimulations showed spontaneous activity (**Figure 11** [↗](#)-supplementary figure 1C) indicating the viability of the nonresponsive cells. Within the pool of responsive cells, the matrix and modules imaged via microprism had higher fractions of Aud-sel cells than those of Som-sel and Aud/Som-nonsel cells (**Figures 11E and G** [↗](#), Chi-Square test, $\chi^2 = 42.5$, $p = 4.4 \times 10^{-8}$), which was similar to the data obtained from the anaesthetized animals. Additionally, matrix had a higher fraction of auditory responsive cells than modules, while modules had a higher fraction of somatosensory responsive cells than matrix (**Figures 11A** [↗](#), D, and H, Chi-Square test, $\chi^2 = 30.2$, $p = 1.2 \times 10^{-6}$). In contrast to the anesthetized animals, the offset cells were very rare (only 6 cells/3 animals), and were evenly distributed between modules and matrix resulting in no significance difference between matrix and modules (**Figures 11C and I** [↗](#), Chi-Square test, $\chi^2 = 2.8$, $p = 0.41$). Similar to anesthetized data, we found that matrix and modules had higher fractions of cells with auditory suppressed responses than those with auditory enhanced responses induced by bimodal stimulation indicating an overall suppression in the cellular auditory response. In contrast to the anesthetized animals, modules had a higher fraction of cells with suppressed response than matrix and a lower fraction of cells with enhanced responses compared to matrix (**Figures 11G and J** [↗](#), χ^2 test = 31.9, $p = 5.2 \times 10^{-7}$) indicating the possibility that the bimodal interactions in modules could be affected by the arousal state of the animal.

Discussion

Summary

As a new imaging approach, a microprism was used to image the LC surface, to provide the first-ever sagittal image of the LC in vivo and to examine its functional organization, which revealed the distinction between strategies used by LC and DC for processing acoustic and non-acoustic information. Imaging the LC either via microprism or from the dorsal surface revealed that it contains GABAergic modules that were embedded in a non-GABAergic matrix. Although modules and matrix were found to be activated by somatosensory and auditory stimuli, they were functionally distinct. The matrix had more auditory-responsive cells with a lower threshold for complex sound detection and had more cells responsive to complex spectral and temporal auditory features than modules. In contrast, modules had more somatosensory responsive cells, consistent with anatomical differences in input to the modules compared to matrix (A. M. Lesicko et al., 2016 [↗](#)). Somatosensory stimulation suppressed auditory responses in the matrix, suggesting that multisensory interactions exist between the matrix and modules.

The novelty of the imaging setup and technical considerations

Microprism-based 2P-microscopy has been successfully used to image the cortical layers in the somatosensory and visual cortices of mice (Andermann et al., 2013 [↗](#)) and to image other inaccessible cortical surfaces like those folded in the brain sulci such as the medial prefrontal cortex (Low, Gu, & Tank, 2014 [↗](#)) or entorhinal cortex (Beckmann et al., 2019 [↗](#); Heys, Rangarajan, & Dombeck, 2014 [↗](#); Low et al., 2014 [↗](#)). In those studies, the glass window glued to the microprism was used, which requires significant skill and potential removal of brain regions to allow prism access (Heys et al., 2014 [↗](#)). Another approach is to use a micro metal or diamond-made knife to create a pocket with minimal tissue damage (Andermann et al., 2013 [↗](#); Marks & Goard, 2021 [↗](#)). Accordingly, we combined and modified some of these procedures to improve the microprism implantation to image the sagittal surface of the LC, which is deeply embedded into the brain tissue in a highly vascular area, with minimum damage to the IC or the surrounding vasculature.

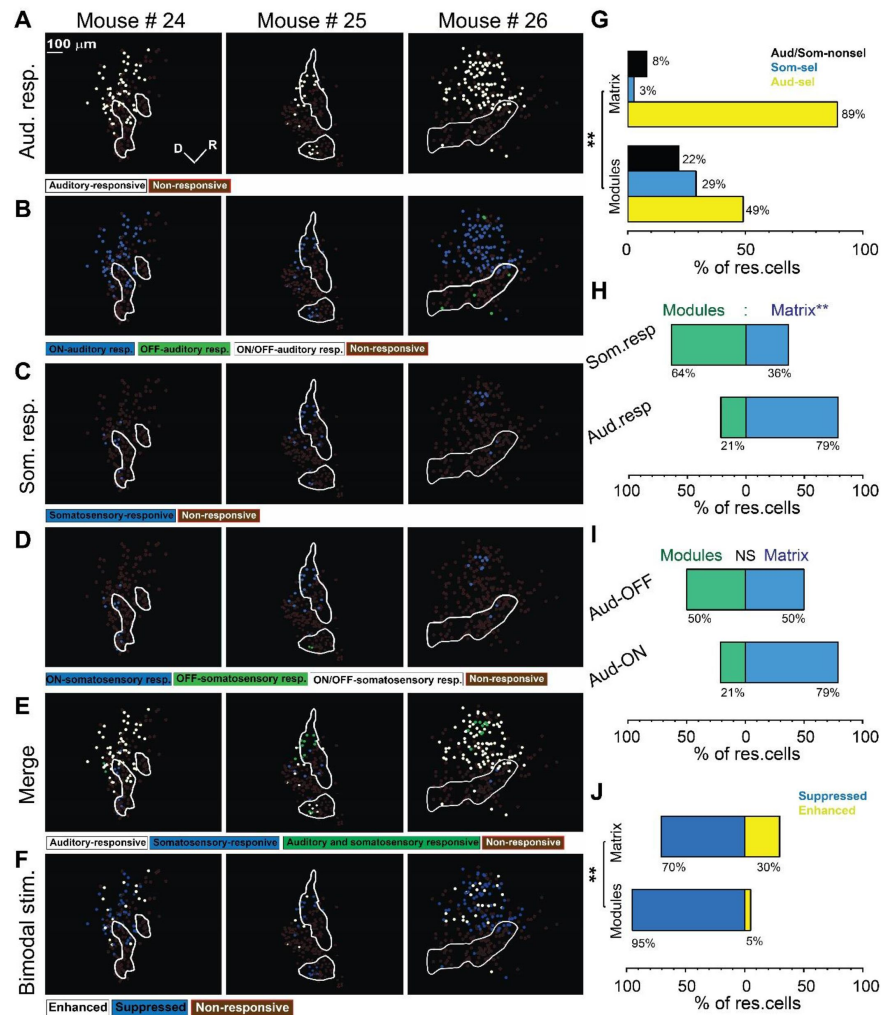


Figure 11

The somatosensory response from LC_(microprism) in awake animals.

(A, C, and E) The pseudocolor images show the auditory (white circles) or somatosensory (blue circles) responsive cells of the LC_(microprism) as well as the merged response (green circles), respectively. (B and D) The pseudocolor images show onset (ON) (blue circles), offset (OFF) (green circles), or onset/offset (ON/OFF) (white circles) responsive cells of the LC_(microprism) to auditory and somatosensory stimuli, respectively. (F) The pseudocolor images show enhanced (white circles) and suppressive response (blue circles) of the auditory responsive cells of the LC_(microprism) following the bimodal stimulation based on their response index (RI). (G) A bar graph showing the percentage of responsive cells to auditory stimulation only or auditory selective cells (Aud-Sel, yellow bars), to somatosensory stimulation only or somatosensory selective cells (Som-sel, blue bars), or nonselectively to both auditory and somatosensory stimulations (Aud/Som-nonsel, black bars) in modules vs matrix (Chi-Square test, $\chi^2 = 42.5$, $p = 4.4 \times 10^{-8}$, % of responsive cells = 49%, 29%, and 22% for Aud-sel, Som-sel, and Aud/Som-nonsel (Modules, 55 cells) and 89%, 3%, and 8% for aud-sel, som-sel, and aud/som-nonsel (matrix, 147 cells) - 3 animals, $**p < 0.05$ vs modules). (H) A bar graph showing the percentage of all cells having auditory responses (Aud. resp) or those having somatosensory responses (Som. resp) in modules vs matrix (Chi-Square test, $\chi^2 = 30.2$, $p = 1.2 \times 10^{-6}$, % of responsive cells (Modules vs matrix) = 21% vs 79% for (aud-resp, 182 cells), 64% vs 36% for (som-resp, 44 cells) - 3 animals, $**p < 0.05$ vs modules). (I) A bar graph showing the percentage of auditory responsive cells with onset (Aud-ON) vs those with offset (Aud-OFF) responses in modules vs matrix (Chi-Square test, $\chi^2 = 2.8$, $p = 0.41$, % of responsive cells (Modules vs matrix) = 21% vs 79% for (aud-ON, 176 cells), 50% vs 50% for (aud-OFF, 6 cells) - 3 animals, $**p < 0.05$ vs modules). (J) A bar graph showing the percentage of auditory responsive cells with suppressed vs those with enhanced responses following bimodal stimulation in modules vs matrix (Chi-Square test = 31.9, $p = 5.2 \times 10^{-7}$, matrix: The percentage of responding cells = 95% vs. 5% for suppressed vs enhanced (Modules, 40 cells) and 70% vs 30% for suppressed vs enhanced (matrix, 142 cells) - 3 animals, $**p < 0.05$ vs modules). D: Dorsal, R: Rostral.

Previous work using microprisms has been done chronically for up to 60 days (Heys et al., 2014; Low et al., 2014; Wenzel, Hamm, Peterka, & Yuste, 2017), with minimal tissue inflammation (Andermann et al., 2013). The current study was initially done on anesthetized animals to examine and validate the new approach, which later revealed similar organizational differences between LC and DC in awake animals. Although imaging through a microprism and 1-mm cover glass appears to have minimal impact on image quality compared to direct 2P imaging (Chia & Levene, 2009), it was important to compare LC imaging via microprism with the direct 2P imaging of the dorsal surface to validate the new imaging method. We found no significant differences between LC response properties when imaging via microprism or imaging directly from the dorsal surface. However, we did observe significant functional differences between the LC and the DC.

Responsiveness to spectral and temporal features

While the LC showed inconsistent and fragmented tonotopy, the DC showed a consistent one, as previously reported (Barnstedt et al., 2015; A. B. Wong & Borst, 2019). These data were supported by a lower R^2 of the best regression fit between BTFs of the LC cells and their locations compared to a higher R^2 obtained for the DC. The direction of the best regression fit shown by the LC was inconsistent across animals, which could be related to the fact that the microprism did not permit a full view of the LC surface. Given that microprism permits imaging of only a single area per animal, it is expected that each animal may have a different profile of cellular organization based on the location of the imaged area on the LC.

The surface of the DC was biased to lower frequencies compared to LC, consistent with previous imaging and recording studies (Ito et al., 2014; Romand & Ehret, 1990; Shen et al., 2003; Stiebler & Ehret, 1985). Despite their response to pure tones, the DC and LC, particularly in the matrix, were more responsive to AM-noise, which fits their function as nonlemniscal divisions (Chen, Cheng, Ito, & Song, 2018; Faye-Lund & Osen, 1985; Loftus et al., 2008).

The spectral and temporal features of animal vocalization and speech are not independent (Singh & Theunissen, 2003). We observed that LC neurons, particularly in the matrix, had greater responsiveness to spectral and temporal modulation than the DC. Given the role of inhibition in spectral and temporal processing (Cai et al., 2018; Chang, Bao, Imaizumi, Schreiner, & Merzenich, 2005; Wehr & Zador, 2003), GABAergic cells of the modules could play an important role in increasing the spectral and temporal modulation indices, especially at lower sound levels, which could be critical for localizing objects in noisy environments given the importance of spectral and temporal cues for sound localization (Davis, Ramachandran, & May, 2003; Goodman, Benichoux, & Brette, 2013; Sinex, 2005). Therefore, future work is critical to image the functional responses of the GABAergic cells of the modules and compare them with the non-GABAergic cells inside and outside the modules.

Somatosensory vs. auditory responses

Consistent with the previous anatomical findings (A. M. Lesicko et al., 2016), modules showed more somatosensory responsive cells than matrix. Since modules receive somatosensory inputs from different brain regions (A. M. Lesicko et al., 2016; Lohse et al., 2022; Zhou & Shore, 2006), the somatosensory response shown by modules could be modulated by different types of somatosensory stimulations, which is an open area for future studies. In contrast to modules, we observed that most auditory-responsive neurons, particularly onset-responsive neurons, were found in the matrix, which receives mostly auditory inputs (A. M. Lesicko et al., 2016). However, a subpopulation of acoustically-driven offset-responsive neurons preferably resides in the modules compared to the matrix. Given that the offset response could be generated as a rebound to the termination of the ongoing inhibition during the sound presentation (Xu, 2014), GABAergic cells in the modules could play an important role in the coding of sound termination as found before in the cortex (Solyga & Barkat, 2021). Because some of the offset acoustically-driven cells

inside the modules were also activated by the somatosensory stimulation, this fraction of cells could then encode the termination of somatosensory-associated sounds through the local inhibitory circuits of the modules. Under in-vivo recording of the mouse IC, the offset responses were previously reported from cells that were non-GABAergic and located in the caudal and shallower parts of the IC (Kasai, Ono, & Ohmori, 2012). These data suggest that the module/matrix system permits the preservation of distinct multimodal response properties, which is consistent with their anatomical distinction, despite the massive integration of inputs in the LC.

Consistent with previous reports (Aitkin et al., 1978; Aitkin, Kenyon, & Philpott, 1981; Jain & Shore, 2006), we observed that combined auditory-somatosensory stimulation generally suppressed neural responses to auditory stimuli and that this suppression was the most prominent in the LC matrix of anesthetized animals. Mechanistically, it is not known if such suppressive responses are created in the LC or are inherited from other brain regions since auditory-somatosensory interactions occur as early as in the cochlear nucleus, and are also found in the IC, thalamus, and cortex (Aitkin et al., 1981; Haenggeli, Pongstaporn, Doucet, & Ryugo, 2005; Kimura, Imbe, Donishi, & Tamai, 2007; Li & Mizuno, 1997; Lohse, Dahmen, Bajo, & King, 2021; Shore, 2005; Shore, Vass, Wys, & Altschuler, 2000; C. Wong, Chabot, Kok, & Lomber, 2015), reviewed in (Brunelle & Llano, 2023; Lohse et al., 2022). We note that a systematic exploration of the stimulus space was not conducted here such that other acoustic or somatosensory stimuli, different relative amplitudes, or timings could have produced enhancement rather than suppression. Future work will be needed to further explore this stimulus space to more fully understand multisensory processing in the LC.

Effect of ketamine anesthesia

The data obtained from anesthetized animals were similar to those obtained from awake animals in the findings related to the tonotopy, the preference of the cells to respond to AM-noise in the LC and DC, the increased sensitivity of matrix to detect the spectral and temporal features of sound, and the bimodal response properties of the matrix vs modules. However, anesthetized and awake animals showed some differences that could be related to the effect of ketamine. For instance, ketamine was found to broaden the RF of the auditory-responsive neurons (Guo et al., 2012). Consistently, the width of the RF of all DC and LC neurons was higher under ketamine anesthesia compared to awake preparation. However, the DC neurons had a wider RF than those in the LC. The relatively narrower RF of the LC could be consistent with their higher SMI (discussed below) compared to that of the DC. Both findings could be related to the degree of inhibition that may shape the inhibitory bands of the RF in the LC neurons given that the LC has prominent GABAergic modules. Also, while the average of the SMIs was negative with no difference between the LC_(microprism) and the DC of awake animals, it was positive with a higher value in the LC_(microprism) compared to the DC of the anesthetized animals indicating the preference of cells to respond more to pure tones under anesthesia. Given that ketamine was reported to increase the duration of the tone-evoked response in the auditory cortex (Guo et al., 2012), the same effect could increase the AUC for the tone-evoked response that might result in a positive average of SMI. Additionally, while LC had a higher average of TMIs than that of the DC of anesthetized animals, the LC had a lower mean of TMIs than that of the DC of awake animals indicating that ketamine may specifically modulate the response properties of the LC via its effects on GABAergic cells (Behrens et al., 2007; Deane et al., 2020; Weckmann et al., 2019), given that GABAergic inhibition reportedly contribute to the processing of the temporally modulated signals (Alluri et al., 2021; Burger & Pollak, 1998; Cai et al., 2018; Gourévitch, Mahrt, Bakay, Elde, & Portfors, 2020). In both anesthetized and awake animals, matrix neurons had always a consistent ratio between the cells with suppressed (~ 70%) and enhanced (~30%) auditory responses induced by bimodal stimulation. However, modules had variable ratios across anesthetized and awake animals. While modules had a lower fraction of cells with enhanced responses than matrix in awake animals, they had a higher fraction of cells with enhanced responses in anesthetized animals, which could be consistent with the direct inhibitory effect of ketamine on GABAergic cells

(Behrens et al., 2007 [↗](#); Deane et al., 2020 [↗](#); Weckmann et al., 2019 [↗](#)). Given that modules are clusters of GABAergic cells and terminals, modules could be specifically modulated by ketamine anesthesia.

Conclusions

Here, using a novel microprism approach to image the LC in anesthetized and awake mice, we observed that 1) DC displayed tonotopic organization which was not consistently present in the LC, 2) DC neurons were generally more sound-responsive than LC neurons but that neurons in both regions had preferential responses to spectrally- and temporally-complex stimuli, 3) that matrix regions of the LC were more responsive to spectrally- and temporally-complex stimuli than modules and that modules were more responsive to somatosensory stimuli and 4) that bimodal stimulation led to suppressive responses that were more prominent in the matrix. In general, these data highlight significant organizational differences between the LC and DC and between modules and matrix in the LC. Future work using microprisms in the LC will help to reveal the potential differential role of matrix vs. module neurons in multisensory integration tasks in behaving animals.

Materials and methods

Animal subjects

Male mice of 8-12 weeks of age were used. Males were used as our preliminary data suggest male/female differences in DC organization (Ibrahim, Shinagawa, Xiao, Asilador, & Llano, 2022 [↗](#)) and will be published separately. GAD67-GFP knock-in mice (developed and shared with permission from Dr. Yuchio Yanagawa at Gunma University and obtained from Dr. Douglas Oliver at the University of Connecticut), where GFP is exclusively expressed in GABAergic cells (Tamamaki et al., 2003 [↗](#)), were used to visualize the GABAergic cells and LC modules. To simultaneously monitor calcium signals and visualize the GFP+ cells, Tg(Thy1-jRGECO1a)GP8.20Dkim/J mice (Jackson Laboratory, Stock# 030525) were crossed with GAD67-GFP knock-in mice to generate GAD67-GFPxThy1-jRGECO1a hybrid mice. The mice were housed and bred at the animal research facility of The Beckman Institute for Advanced Science and Technology. The animal care facilities are approved by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC). To phenotype the GFP-positive animals, a single-photon fluorescence microscope was used for the transcranial examination of the pups at postnatal day 4 using excitation: 472/30 nm and emission 520/35 nm filters along with dichroic 505 nm long pass at low power magnification (2.5×/0.08, Olympus objective, MPlanFL N, Japan). The positive pups exhibited a green fluorescence in the cerebellum, the cerebral cortex, and the olfactory bulb and were only kept with their parents until weaning. The genotyping for jRGECO1a was done using a probe composed of a forward primer: GCCGCCGAGGTCAAGA & a reverse primer: TCCAACCTTGATGTCGACGATGTAG by Transnetyx™ (Transnetyx, USA). Samples of the mice's tail snips around one month of age were used for genotyping. All applicable guidelines for the care and use of animals were followed. All procedures were approved by the Institutional Animal Care and Use Committee (IACUC).

Surgery

The general procedures of craniotomy were previously described (Goldey et al., 2014 [↗](#)), with modifications to place craniotomy over the IC. Before surgery, mice were anesthetized with a mixture of ketamine, xylazine, and acepromazine (100 mg/kg, 3 mg/kg, and 3mg/kg, respectively) delivered intraperitoneally. The anesthesia was maintained during the surgery and imaging using only ketamine (100 mg/kg). To prevent neural edema during or after the craniotomy, an

intramuscular injection of dexamethasone sodium (4.8 mg/kg) was given just before the surgery using an insulin syringe. After placing the animal in the stereotaxic apparatus (David Kopf Instruments, USA), both eyes were protected by applying Optixcare lubricant eye gel (Aventix Animal Health, Canada). The hair on the scalp was then removed by massaging the scalp with a depilatory cream (Nair) using a cotton-tipped applicator and leaving the cream on the scalp for 4-5 minutes. The cream was then removed by a thin plastic sheet (flexible ruler) to leave a hair-free area on the scalp. The remaining tiny hairs were then removed by alcohol swab and the area was then sterilized by applying 10% Povidone Iodine (Dynarex, USA) using a sterile cotton-tipped applicator. The medial incision was made with a scalpel blade #10, and 0.2 ml of 0.5% lidocaine was injected intradermally into the scalp. The skin extending from the medial line to the temporalis muscle was completely removed using a pair of microscissors to produce a wide skinless area above the skull. Number 5/45 forceps were used to remove any remaining periosteum. The remaining dried or firmly attached pieces of periosteum were removed with a scalpel blade #10. The skull was cleaned with sterile saline and dried with gently pressurized air. Using the stereotaxic apparatus, a wide area of ~3 x 4 mm above the left IC was made. A micro drill bit (size #80, Grainger, USA) was used to drill through the skull starting from the rostralateral (Figure 1A, yellow circle) region to lambda (Figure 1A, black circle) following the border in a clockwise direction (Figure 1A, dotted black rectangle). To prevent overheating of the superficial brain tissues and to mitigate the occasional spurts of skull bleeding during the drilling, ice-cold sterile saline was used to intermittently irrigate the surface. A stream of pressurized air was also applied during the drilling procedures to prevent overheating and remove the debris produced by the drilling. Caution was taken not to pierce the dura when performing the craniotomy while crossing the sagittal or the lambdoid sutures to avoid damaging the underlying sinuses. After drilling, the skull was irrigated in sterile saline and the bone flap (the undrilled bone over the craniotomy area) was gently examined for complete separation from the rest of the skull (Figure 1B). Using a pair of no. 5/45 forceps, the bone flap was gently removed. To control the bleeding if it occurred, a piece of sterile hemostatic gel (Adsorbable Gelatin Sponge USP, Haemosponge, GBI, India) which was pre-soaked in ice-cold saline, was applied to the bleeding spot. Once the bleeding ceased, the brain was kept covered in sterile saline. In some surgeries, the dura was peeled off from the surface of the IC while removing the bone flap. In the surgeries where the dura remained intact, a Bonn microprobe (F.S.T, Germany, Item # 10032-13) was used to pierce the dura in an area that is not above the IC and devoid of cortical vasculature (e.g., a part of exposed cerebellum). After piercing, the dura was carefully and gently lifted, and a pair of no. 5/45 forceps were used to grab the dura to gently tear it to the extent of the transverse sinus to avoid bleeding of the major venous structures.

Using the stereotaxic apparatus, a 30° diamond knife (F.S.T, Germany, Item # 10100-30) was placed in its vertical position. From the medial line, the knife was placed at the lateral horizon of the IC. The knife was then inserted 1.5 mm ventrally. Caution was taken to avoid hitting any major blood vessels to avoid bleeding. From its position, the knife was retracted 0.3-0.5 mm medially to create an initial pocket for the insertion of the microprism (Figures 1C-E). The microprism (1.5 mm-silver coated, Tower Optical Corporation) was then inserted, so one of its sides was parallel to the lateral surface of the diamond blade, the other side was facing up, and its hypotenuse was facing ventrally (Figures 1F and G). The wooden piece of a cotton swab was carefully driven down to push the microprism until its top surface was at the same level as the IC dorsal surface. Keeping the microprism in its place with the wooden end of a swab, the diamond knife was carefully retracted upward leaving the microprism behind (Figure 1H). If the retraction of the diamond knife pulled the prism up, the wooden piece would be used to push it down simultaneously with the knife movement. To control the bleeding, a piece of sterile hemostatic gel was used as before. The tip of the diamond knife was used to secure the microprism in its position for at least 10 minutes (Figure J). During that time, the solution of the 1% agarose was prepared for use. A pair of forceps were then used to gently place a sterile 5 mm cover glass #1 (Thomas Scientific, USA) over the skull covering the opened area above the IC. The cover glass was secured by a wooden trimmed piece of sterile cotton swab by gently pressing the cover glass from the top. Since the

surface of the IC was located below the surfaces of both the cerebellum and cerebral cortex, the cover glass usually traps about a 0.7-1.0 mm thick sheet of sterile saline. The movement of the liquid medium caused by heart pulsation was found to be a source of motion artifact during imaging, so the gap between the surface of the IC and the cover glass was filled with 1% agarose gel that was made in saline at a temperature range of 33-35°C (**Figure 1K**). The extra agarose gel pieces were cut off and removed by the scalpel blade and the wet skull was then dried out using pressurized air. A titanium headpost as described before (Goldey et al., 2014) was glued carefully on the top of the skull to be at the same level as the cover glass. Following the manufacturer's instructions, the C&B Metabond (Parkell, Japan) was used to secure the headpost in its place. Excluding the steps of the microprism insertion, the same procedures were followed for DC imaging from the surface (**Figure 1L**). For awake preparations, the same procedures were followed under isoflurane anesthesia starting with 4% of isoflurane as an induction dose with 1-2% isoflurane during the surgery. The animal was transferred under the microscope objective and left to recover from isoflurane anesthesia for 1 hour before imaging. To ensure the animal's recovery, the animal was visually inspected for its breathing rate (Ewald, Werb, & Egeblad, 2011) and ongoing limb movements.

Acoustic stimulation

Using a custom-made MATLAB (The MathWorks, Natick, MA, USA) code, either 500 ms pure tone or AM-noise was generated. Thirty-five (5×7) pure tones with a series of sound pressure levels (40, 50, 60, 70, 80 dB SPL) and a series of carrier frequencies (5000-40000 Hz with a half-octave gap) were presented with a cosine window. Forty-five (5×9) 100% AM white noise bursts with a series of sound pressure levels (40, 50, 60, 70, 80 dB SPL) and modulation frequencies (0, 2, 4, 8, 16, 32, 64, 128, 256 Hz) were generated. The stimuli of either pure tone or AM-noise combinations were played in random sequence to the mice with a 600 ms interstimulus interval (ISI) by a TDT RP2.1 processor (Tucker-Davis Technologies, US) and delivered by a TDT ES1 speaker (Tucker-Davis Technologies, US). In other sets of experiments, 500 ms of unmodulated broadband noise was presented 5 times with ISI of 600 ms at 80 dB SPL to examine the acoustic activity between modules and matrix compared to somatosensory stimulation.

The output of the TDT ES1 speaker was calibrated using a PCB 377A06 microphone, which feeds a SigCal tool to generate a calibration file for all tested frequencies (5–40 kHz). To enable the custom-made MATLAB code to read this calibration file, the values were first processed by MATLAB signal processing toolbox (sptool) to generate a 256-tap FIR filter to apply the calibration using the following parameters [arbitrary magnitudes, least square, order: 256, sampling rate: 97,656.25, frequency vector (5–40 kHz), amplitude vector (40–80 dB SPL), and weight vector ones (1128)].

Somatosensory stimulation

The somatosensory stimulation was done by deflecting the right whiskers of the animals at an average excursion of 1.2 mm with a 50 Hz rate over 500 ms to match the acoustic stimulus. A total of five stimuli were presented with 600 ms ISI. The whisker deflection was performed using a Brüel & Kjær vibrator (V203 10/32 UNF-CE, United Kingdom) that was driven by a wave generator (4063, BK precision, USA) after amplifying the signals with an SLA1 100 W power amplifier (ART, USA).

Two-Photon imaging

Immediately after surgery, the anesthetized animal was taken and secured under the microscope objective by clamping the arms of the head post to two perpendicular metal posts mounted on the microscope stage. A custom-built 2P microscope was used. The optical and the controlling components were supplied from Bruker, Olympus, and Thorlabs. The imaging of the DC was made using a 20x water-immersion objective (LUMPlanFI/IR, 20X, NA: 0.95, WD: 2 mm; Olympus Corporation, Tokyo, Japan), while the lateral surface of the IC was imaged by the long working

distance 16x water-immersion objective (N16XLWD-PF - CFI LWD Plan Fluorite Objective, 16X, NA: 0.8, WD: 3.0 mm; Nikon, Tokyo, Japan) to be able to reach the focal point of the microprism at its hypotenuse. For imaging both the GFP or jRGECO1a signals, the excitation light was generated by InSight X3 laser (Spectra-Physics Lasers, Mountain View, CA, USA) tuned to a wavelength of 920 or 1040 nm, respectively. A layer of a 1:1 mixture of wavelength multipurpose ultrasound gel (National Therapy, Canada) with double-deionized water was used to immerse the objective. This gel was able to trap the water and reduce its evaporation during imaging. The emitted signals were detected by a photomultiplier tube (Hamamatsu H7422PA-4, Japan) following a t565lp dichroic and a Chroma barrier et525/70m filter for GFP and et595/50m filter for jRGECO1a signals. Images (512x512 pixels) were collected at a frame rate of 29.9 Hz in the resonant galvo mode. Data were collected from the dorsal surface of the IC by scanning the surface of the IC based on the GFP and jRGECO1a signals through the medial and lateral horizons of the IC. Generally, the scanning was started by moving the 20x objective to the most ventromedial position, where there is no ability to see any cells (**Figure 2B**, medial horizon). Then, the objective was driven to slowly move dorsolaterally to get cells in focus (**Figure 1M**), where the average depth of the first field of view was 169 μm . The lateral movement of the objective was intended to be aligned and parallel to the curvature of the DC surface indicated by getting the superficial cells in focus to avoid any possibility of imaging the upper layers of the central nucleus of the IC, so the average depth of the most superficial point of the imaging was 92 μm . The scanning was ended by moving the objective to the most ventrolateral position of the IC, where there is no ability to see GFP laterally (**Figure 2B**, lateral horizon), where the last field of view was collected at an average depth of 225 μm . Generally, each field of view was selected based on the expression of GFP signals as an indicator for the cells in focus and being acoustically active by using a search stimulus that was 500 ms broadband noise with zero modulation at 80 dB SPL. The z-axis of the objective was determined relative to the pia surface. The relative position of each region was tracked through the micromanipulator (MP-285, Sutter Instruments) that controlled the microscope objective. The positions of imaged areas were further aligned across animals to a common coordinate using the midline, lateral extremes, and the vasculature landmarks taken via imaging of the GFP signals of the cranial window with low magnification (4x Olympus objective, 4x4 binning, 50 ms frame time) using a Lumen 200 bulb (Prior Scientific Inc, USA) and CoolSNAP MYO camera (Photometrics, USA, excitation: 488 nm and emission: 515–550 nm). Imaging of the lateral surface of the IC via microprism was done similarly. In both cases, the frame timing of the scanner and the sound stimuli were both digitized and time-locked using a Digidata 1440A (Molecular Devices, Sunnyvale, CA, USA) with Clampex v. 10.3 (Molecular Devices, Sunnyvale, CA, USA).

Laser lesion

The laser lesion was produced using a 720 or 820 nm laser beam at 74-80 mW power at the level of the objective for 100 to 250 seconds depending on the size of the lesioned area, the speed of laser scanning (frame rate), and the depth of the lesion (z-axis length). In brief, the area of interest and the dimension of the laser lesion were initially determined using 920 nm as indicated by the GFP signals through scanning one of the GABAergic modules. The depth of the GABAergic modules was then determined by counting the optical slices starting from their superficial surface. The tunable laser was then switched from 920 nm to the shorter wavelength (720 or 820 nm) to scan the same area and depth at a rate (1-2 frames/ second).

Data processing

Data collection

The data were collected as separate movies (512x512 pixels) for each pure tone or AM-noise runs in a resonant galvo mode. Depending on the amplitude and frequency combinations for each type of acoustic stimulus, 40- or 52-second periods were assigned as a movie's length for pure tone (35 stimulus combinations), or AM-noise (45 stimulus combinations), respectively. Using ImageJ

software, the z-projection was used to compute one single image representing either the sum, the standard deviation, or the median of all the image sequences in the movie. Based on these single images, the region of interest (ROI) was manually drawn around each detectable cell body. **Figures 3A and B** represent the sum of the image sequences of the movie collected from the dorsal surface, or the lateral surface of the IC via microprism, respectively. For the subsequent processing steps, Python open-source libraries were used (listed below).

Motion Correction and Filtering

The imread function from the OpenCV library was used in grayscale mode to import images into a numpy array from a single folder containing TIFF images. The array was then exported as a single TIFF stack into a temporary folder using the mimwrite function from the imageio library, and the process was repeated for each folder. The NoRMCorre algorithm (Pnevmatikakis & Giovannucci, 2017) embedded in the CaImAn library (Giovannucci et al., 2019) was used to apply motion correction to each of the TIFF files. The data were then imported into a numpy array, rounded, and converted to 16-bit integers. The images were filtered using a 2D Gaussian filter with a sigma value of 1 (surface View) / 2 (prism View) in each direction, then a 1D Gaussian temporal filter with a sigma value of 2 was applied using the ndimage.gaussian_filter and ndimage.gaussian_filter1d function from the scipy library, respectively.

Data Extraction

The ROI sets, which were manually created using ImageJ, were imported using the read_roi_zip function from the read_roi library. The sets were then used to create two masks; one mask was used as a replica of the ROIs and the second mask was made around the original ROI (roughly four times larger in area). The smaller mask was applied to find the average pixel value within each ROI, while the larger mask was applied to find the average pixel value of the neuropil. The neuropil correction was applied using the following equation (Akerboom et al., 2012; Kerlin, Andermann, Berezovskii, & Reid, 2010);

$$\text{Corrected value} = \text{Date value} - (0.4 \times \text{Neuropil value})$$

To identify the calcium signals, $\Delta f/f$ was calculated by using the following equation.

$$\frac{\Delta f}{f} = \frac{(\text{Date value} - \text{Background value})}{\text{Background value}}$$

where the background value is the slope estimating the background levels with fluorescence bleaching factored in. The data were then reorganized so that all segments with the same stimulus frequency and stimulus amplitude were grouped. The area under the curve (AUC) of the calcium signals of the excitatory responses only was then used as a metric to determine the magnitude of the response and for subsequent analysis (Berens et al., 2018; Overk, Rockenstein, Florio, Cheng, & Masliah, 2015).

Cell Flagging

The correlation coefficient between each of the trials was calculated using the stats. Pearson function from the scipy library. The average correlation coefficient was calculated for each stimulus frequency and stimulus amplitude. Similar to previous work (A. B. Wong & Borst, 2019), if the average correlation coefficient was above the threshold of 0.6, the cell was flagged as being responsive to that stimulus combination of frequency and amplitude, and the best tone

and modulation frequencies were calculated for every cell (Barnstedt et al., 2015 [↗](#)). Knowing BTF and the best modulation frequency (BMF) for each neuron enabled us to calculate the SMI and TMI at each sound level following these equations;

$$\text{SMI}$$

$$= \frac{\text{Response to BTF} - \text{Response to unmodulated noise}}{\text{Response to BTF} + \text{Response to unmodulated noise}}$$

The SMI has values ranging from (-1 to 1). While neurons with values closer to 1 are more responsive to pure tone, neurons with values closer to -1 are more responsive to noise with no modulation.

$$\text{TMI}$$

$$= \frac{\text{Response to BMF} - \text{Response to unmodulated noise}}{\text{Response of BMF} + \text{Response to unmodulated noise}}$$

Similarly, the TMI has values ranging from (-1 to 1). While neurons with values closer to 1 are more responsive to AM-noise, neurons with values closer to -1 are more responsive to noise with no modulation.

For the comparison between somatosensory and auditory responses, the threshold of 0.4 was used as a cutoff threshold to determine the responsive cells to each stimulation. The average response of each somatosensory, acoustic (500 ms of unmodulated noise), or simultaneous somatosensory and acoustic (bimodal) stimulation was then calculated for each responsive cell. The correlation method was validated as a tool to determine the responsive cells to somatosensory stimulation by generating an ROC curve comparing the automated method to a blinded human interpretation. The AUC of the ROC curve was 0.88. This high AUC value indicates that the correlation method can rank the random responsive cells than the random nonresponsive cells. At the correlation coefficient (0.4), which was the cutoff value to determine the responsive cells for somatosensory stimulation, the specificity was 87% and the sensitivity 72%, the positive predictive value was 73%, and the negative predictive value was 86%. Additionally, all the false-responsive cells were excluded from the analysis. To examine the effect of somatosensory stimulation on auditory responses, the response index (RI) was calculated for the auditory-responsive cells using the following equation;

$$\text{RI}$$

$$= \frac{\text{Auditory response to bimodal stimulation}}{\text{Auditory response to auditory stimulation only}}$$

Neurons with RI values > 1 have enhanced or higher responses induced by simultaneous auditory and somatosensory stimulation, while neurons with RI values < 1 have downregulated or suppressed auditory responses induced by auditory and simultaneous somatosensory stimulation.

Map Generation

The average radius for all ROIs was calculated to ensure that all cells on the tonotopic map had uniform radii. A color key was also generated for every corresponding map. A blank canvas was generated using the Image module from the pillow library, and a circle was created for each cell and filled with a corresponding color to its value.

Receptive field sum (RFS) analysis

The binary RFS was calculated for every cell by determining whether the cell was responsive to a particular combination of frequency and amplitude (Bowen et al., 2020 [DOI](#)). A value of 1 was given to the responsive cell, if it responds to a single frequency/amplitude combination. In contrast, a value of 1 was added to the tally for each responsive combination if the cell was broadly responsive. Accordingly, the higher and lower values of RFS indicate a broad and narrow RF, respectively. To examine the differences between the RFS of the cells at different areas under different preparations, the cumulative distribution function was used.

Assessment of tonotopic organization

The tonotopy was assessed by examining the best fit of linear or nonlinear quadratic polynomial regressions between the BTFs of cells and their locations along different anatomical axes. For the LC_(microprism), the regression fit was examined along the main four axes such as dorsal to ventral, rostral to caudal, dorsocaudal to ventrostral, and dorsorostral to ventrocaudal. Based on the previous reports showing the tonotopic organization over the DC (Barnstedt et al., 2015 [DOI](#); A. B. Wong & Borst, 2019 [DOI](#)), the tonotopy was examined along the rostromedial to caudolateral axis. The x_2 and y_2 coordinates of each cell were determined. At the starting point of each axis, the x_1 and y_1 coordinates were assigned values as (0, 0). Then the distance (d) between each cell and the starting point at each axis was calculated based on the following equation;

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For each 50 μm (bin size), the geometric average of the BTFs of all cells as well as their distances from the starting point at each axis were computed. The linear or quadratic polynomial regressions were then examined between the geometric averages of BTFs and the geometric averages of distances of cells for each animal.

Detecting the spontaneous activity from the nonresponsive cells

Movies have been recorded from the nonresponsive cells of the LC_(microprism) when no stimulation was presented. The time trace of jRGECO1a signals was extracted from each cell. A polynomial fitting of the time trace was used to correct for photobleaching and to adjust for the baseline for the whole trace. The sliding average of the baseline was then calculated for 1 second (~29 frames), and the z score was calculated for the subsequent frames. The spontaneous activity was detected when the z-score was > 3.0.

Statistics

All statistical analyses and graphs were made using Origin Pro software. The normality of the distributions of the data was examined using a Kolmogorov-Smirnov test. The Mann-Whitney test as well as two-sample or paired t-test were used to examine the difference in data distribution or the difference between samples of non-normal and normal distribution, respectively. Either one-way or two-way ANOVA as well as Fisher as a post hoc test were used for the mean comparison between different groups of normal distribution. The Chi-square (χ^2) test or Kruskal-Wallis ANOVA tests were used to compare the samples based on cell numbers of their cellular population. The cumulative frequency function was computed to calculate the fraction of cells tuned at each frequency based on their best frequency. Differences were deemed significant when p-value < 0.05. The upper and lower borders of each box plot are at the 25th and 75th percentile, with whiskers at 95% spread.

Histology

After the 2P imaging, the mouse was transcardially perfused with 4% paraformaldehyde (PFA) solution in PBS solution. The isolated brain was then placed overnight in PFA for a complete fixation. The brain was then moved from the PFA to a graded sucrose solution (10, 20, and 30%) until it sunk. 50 μm coronal sections were taken at the level of the IC using a Leica cryostat. The sections were then mounted on a slide, coverslipped, and imaged using a confocal microscope for imaging the GFP signals as an indicator for the GABAergic cells in the GAD67-GFP knock-in mouse (excitation: 488 nm and emissions: 515–550 nm).

For the examination of *jrgeco1a* expression in the IC, the mouse was transcardially perfused with saline with a high concentration of potassium and a high ratio of calcium to magnesium [(in mM) 3.5 KCl, 25 glucose, 125 NaCl, 1.2 KH_2PO_4 , 2 CaCl_2 , 0.5 MgSO_4] under ketamine anesthesia. The brain was then fixed by a subsequent perfusion of 4% PFA. Selected 40 μm coronal sections at the level of the IC were immunostained for NeuN using (NeuN (D4G40) XP® Rabbit mAB, Cell Signaling, Cat# 62994, 1:100) primary antibody and (goat anti-rabbit IgG conjugated to Alexa 405, 1:200) secondary antibody following the immunostaining procedures that were previously described (Issa, Sekaran, & Llano, 2023 [\[1\]](#); Vaithiyalingam Chandra Sekaran et al., 2021 [\[2\]](#)). Confocal images were obtained using a Confocal Zeiss LSM 710 Microscope using 10 and 40x objectives and tile scan functions. For NeuN, Alexa 405 signals were obtained by a 405 nm laser and emission at 415–466 nm. The GFP signals were obtained by a 488 nm laser and emission at 501–551 nm. The *jRGECO1a* signals were obtained by 561 nm laser and emission at 600–700 nm. All cells with somas expressing a red fluorescence of *jrgeco1a* were quantified in 2P-imaged areas of the DC and LC. The density of the cells expressing *jrgeco1a* was then calculated based on the cells expressing NeuN.

Artwork

The cartoon of the mouse brain in **Figures 1** [\[1\]](#) C, F, and I was obtained from Brain Explorer 3D viewer Allen Mouse Brain (<https://connectivity.brain-map.org/3d-viewer?v=1> [\[3\]](#)) and modified to be suitable for illustration. All figures were designed and made using Adobe Illustrator (Adobe, San Jose, CA). To keep working within the Adobe environment to avoid losing the resolution of the figures, Adobe Photoshop (Adobe, San Jose, CA) was used to crop the borders of some images to save space. The color pallets were chosen to avoid the combination of the green and red colors for all maps, so they will be suitable for color-blind readers. To induce the same effect in the microscope images combining the red (*jRGECO1a*) and green (GFP) fluorescence, we kept the green signals of the GFP unchanged, but we substituted the red fluorescence of *jRGECO1a* with magenta as a pseudocolor.

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Competing interests

On behalf of all authors, there are no financial or non-financial competing interests to be declared.

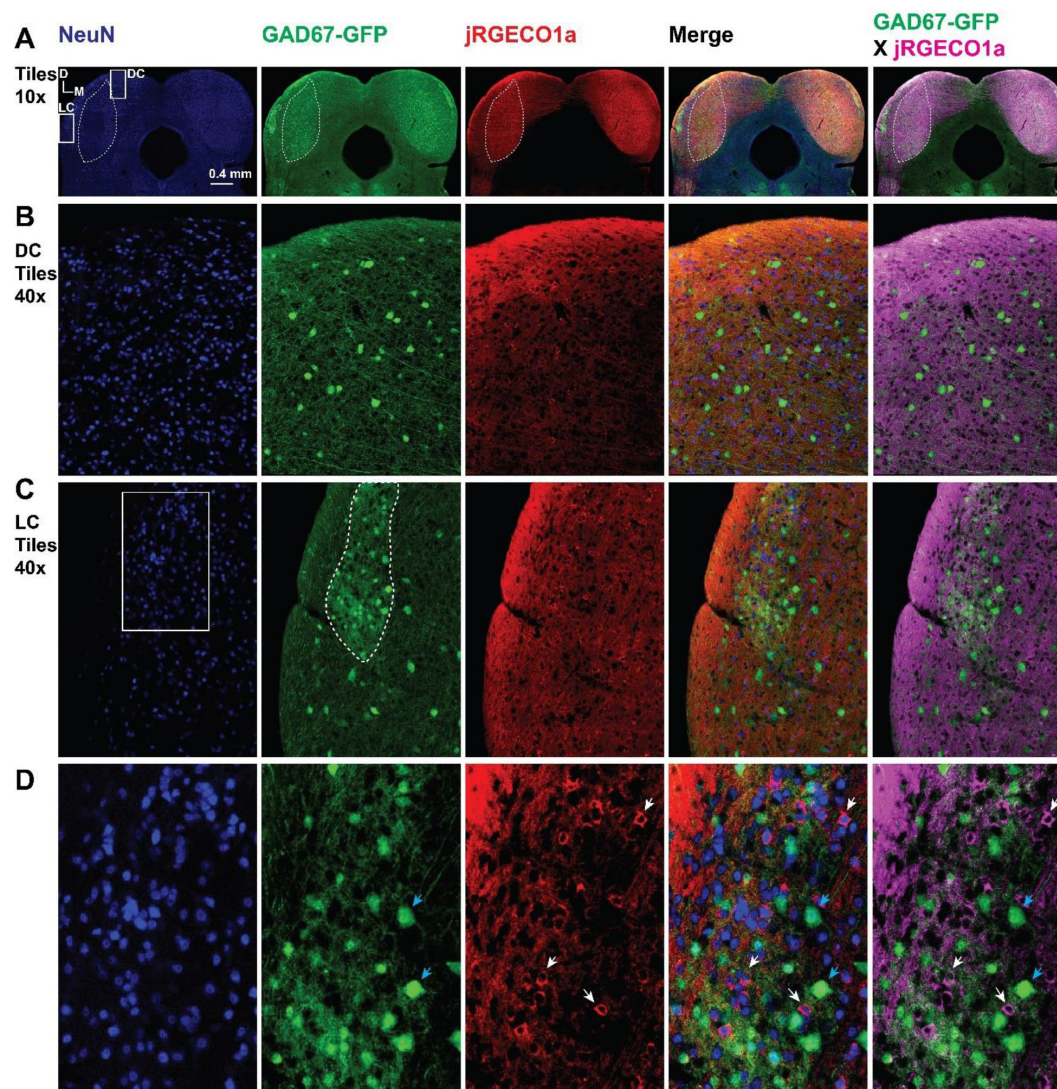


Figure 3-figure supplement 1

(A) Image of the IC showing the immunostaining to NeuN (blue-column 1), GAD67-GFP (green-column 3), jRGECO1a (red-column 3), merged image for the three proteins (column 4), and merged image for jRGECO1a (magenta) and GFP (green) – dotted lines represents the central nucleus of the IC in all channels as well as the modules in the green channel. (B) A magnified view of the DC in A – dotted line represents the central nucleus of the IC (Icc). (C) A magnified view of the LC in A – dotted line represents one of the GABAergic modules of the LC. (D) A magnified view of the white square in C shows some GABAergic cells (blue arrows) and some nonGABAergic cells expressing jRGECO1a (white arrows). D: dorsal, L: lateral.

Figure 4-figure supplement 1

(A-B) Linear and nonlinear quadratic regression fit between the BTFs and the locations of the cells of the LC_(microprism) and the DC, respectively.

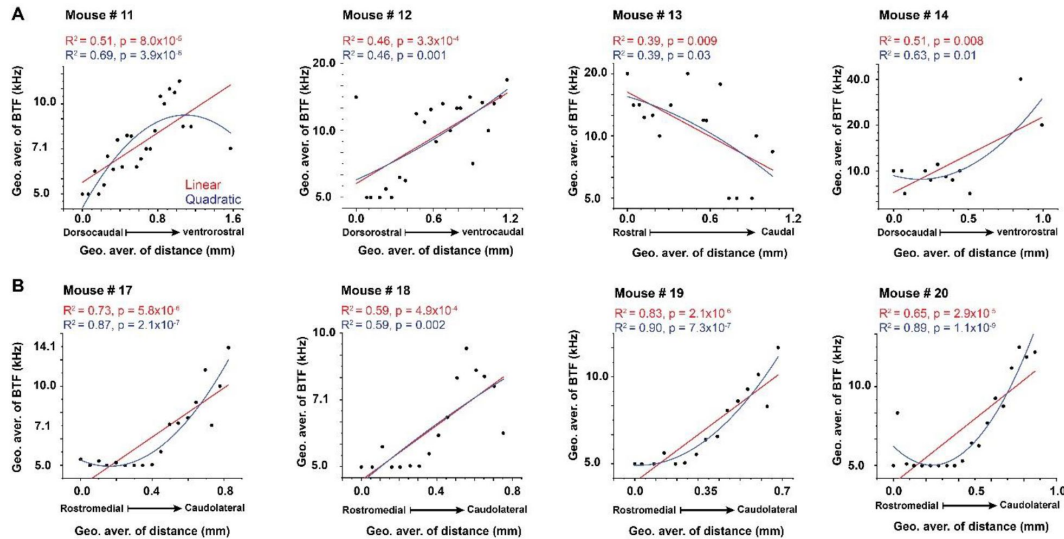
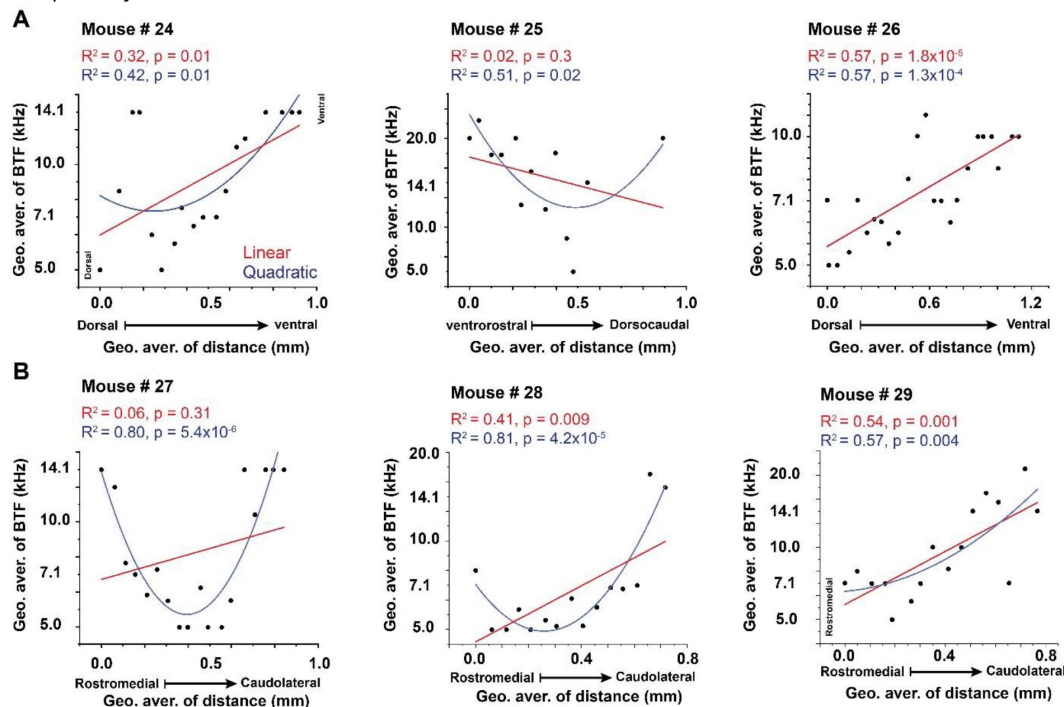


Figure 9-figure supplement 1

(A-B) Linear and nonlinear quadratic regression fit between the BTFs and the locations of the cells of the LC_(microprism) and the DC, respectively.



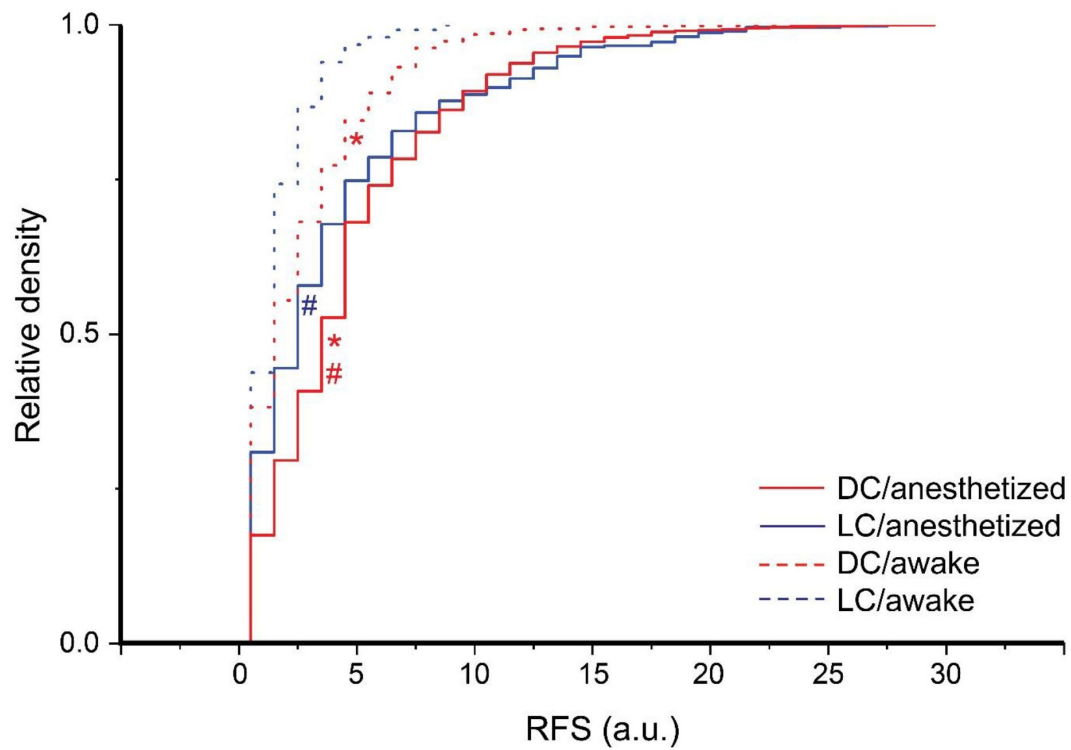


Figure 10-figure supplement 1

The RFS of DC and LC neurons at different experimental preparations:

The cumulative distribution function of the RFS of all cells collected from DC or LC under different experimental preparations (Kruskal-Wallis ANOVA, $\chi^2 = 346.9$, $p < 0.001$, posthoc Dunn's test; *(DC vs LC): $z = -6.4$ and -4.47 with $p < 0.001$ and 4.6×10^{-5} under anesthesia and awake preparations, respectively; #(anesthesia vs awake): $z = 13.6$ and 7.8 with $p < 0.001$ for DC and LC, respectively; mean $\pm$ SEM = 5.1 ± 0.08 , 4.5 ± 0.2 , 3.07 ± 0.1 , and 2.0 ± 0.08 for (DC/anesthesia, median = 4, 2487 cells / 4 animals), (LC/anesthesia, median = 3, 472 cells / 7 animals), (DC/awake, median = 2, 647 cells / 3 animals), (LC/awake, median = 2, 249 cells / 3 animals).

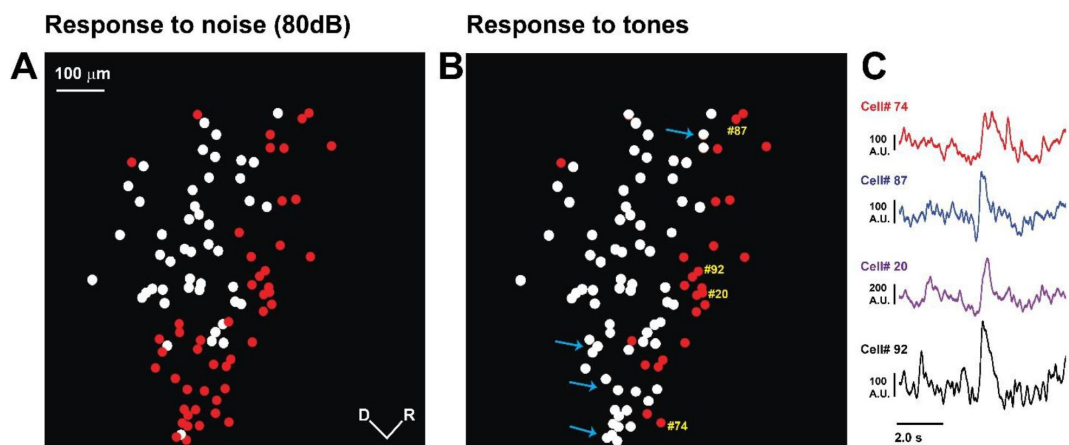


Figure 11-figure supplement 1

The activity of the nonresponsive cells of the LC_(microprism) in awake animals.

(A) The pseudocolor image shows the nonresponsive (red circles) vs responsive (white circles) cells to the auditory stimulation with unmodulated broadband noise at the LC_(microprism). (B) The pseudocolor image shows the nonresponsive (red circles) vs responsive (white circles) cells to the auditory stimulation with unmodulated broadband noise or pure tones (blue arrows) at the LC_(microprism). (C) Time traces of the spontaneous activity of four nonresponsive cells (yellow numbers in B).

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Reviewer #1 (Public Review):

In this paper the authors provide a characterisation of auditory responses (tones, noise, and amplitude modulated sounds) and bimodal (somatosensory-auditory) responses and interactions in the higher order lateral cortex (LC) of the inferior colliculus (IC) and compare these characteristic with the higher order dorsal cortex (DC) of the IC - in awake and anaesthetised mice. Dan Llano's group have previously identified gaba'ergic patches (modules) in the LC distinctly receiving inputs from somatosensory structures, surrounded by matrix regions receiving inputs from auditory cortex. They here use 2P calcium imaging combined with an implanted prism to - for the first time - get functional optical access to these subregions (modules and matrix) in the lateral cortex of IC in vivo, in order to also characterise the functional difference in these subparts of LC. They find that both DC and LC of both awake and anaesthetised appears to be more responsive to more complex sounds (amplitude modulated noise) compared to pure tones and that under anesthesia the matrix of LC is more modulated by specific frequency and temporal content compared to the gaba'ergic modules in LC. However, while both LC and DC appears to have low frequency preferences, this preference for low frequencies is more pronounced in DC. Furthermore, in both awake and anesthetized mice somatosensory inputs are capable of driving responses on its own in the modules of LC, but very little in the matrix. The authors now compare bimodal interactions under anaesthesia and awake states and find that effects are different in some cases under awake and anesthesia - particularly related to bimodal suppression and enhancement in the modules.

The paper provides new information about how subregions with different inputs and neurochemical profiles in the higher order auditory midbrain process auditory and multisensory information, and is useful for the auditory and multisensory circuits neuroscience community.

The manuscript is improved by the response to reviewers. The authors have addressed my comments by adding new figures and panels, streamlining the analysis between awake and anaesthetised data (which has led to a more nuanced, and better supported conclusion), and adding more examples to better understand the underlying data. In streamlining the analyses between anaesthetised and awake data I would probably have opted for bringing these results into merged figures to avoid repetitiveness and aid comparison, but I acknowledge that that may be a matter of style. The added discussions of differences between awake and anaesthesia in the findings and the discussion of possible reasons why these differences are present help broaden the understanding of what the data looks like and how anaesthesia can affect these circuits.

As mentioned in my previous review, the strength of this study is in its demonstration of using prism 2p imaging to image the lateral shell of IC to gain access to its neurochemically defined subdivisions, and they use this method to provide a basic description of the auditory and multisensory properties of lateral cortex IC subdivisions (and compare it to dorsal cortex of IC). The added analysis, information and figures provide a more convincing foundation for

the descriptions and conclusions stated in the paper. The description of the basic functionality of the lateral cortex of the IC are useful for researchers interested in basic multisensory interactions and auditory processing and circuits. The paper provides a technical foundation for future studies (as the authors also mention), exploring how these neurochemically defined subdivisions receiving distinct descending projections from cortex contribute to auditory and multisensory based behaviour.

Minor comment:

- The authors have now added statistics and figures to support their claims about tonotopy in DC and LC. I asked for and I think allows readers to better understand the tonotopical organisation in these areas. One of the conclusions by the authors is that the quadratic fit is a better fit than a linear fit in DCIC. Given the new plots shown and previous studies this is likely true, though it is worth highlighting that adding parameters to a fitting procedure (as in the case when moving from linear to quadratic fit) will likely lead to a better fit due to the increased flexibility of the fitting procedure.

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Reviewer #2 (Public Review):

Summary:

The study describes differences in responses to sounds and whisker deflections as well as combinations of these stimuli in different neurochemically defined subsections of the lateral and dorsal cortex of the inferior colliculus in anaesthetised and awake mice.

Strengths:

A major achievement of the work lies in obtaining the data in the first place as this required establishing and refining a challenging surgical procedure to insert a prism that enabled the authors to visualise the lateral surface of the inferior colliculus. Using this approach, the authors were then able to provide the first functional comparison of neural responses inside and outside of the GABA-rich modules of the lateral cortex. The strongest and most interesting aspects of the results, in my opinion, concern the interactions of auditory and somatosensory stimulation. For instance, the authors find that a) somatosensory-responses are strongest inside the modules and b) somatosensory-auditory suppression is stronger in the matrix than in the modules. This suggests that, while somatosensory inputs preferentially target the GABA-rich modules, they do not exclusively target GABAergic neurons within the modules (given that the authors record exclusively from excitatory neurons we wouldn't expect to see somatosensory responses if they targeted exclusively GABAergic neurons) and that the GABAergic neurons of the modules (consistent with previous work) preferentially impact neurons outside the modules, i.e. via long-range connections.

Weaknesses:

While the findings are of interest to the subfield they have only rather limited implications beyond it and the writing is not quite as precise as it could be.

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Reviewer #3 (Public Review):

The lateral cortex of the inferior colliculus (LC) is a region of the auditory midbrain noted for receiving both auditory and somatosensory input. Anatomical studies have established that somatosensory input primarily impinges on "modular" regions of the LC, which are

characterized by high densities of GABAergic neurons, while auditory input is more prominent in the "matrix" regions that surround the modules. However, how auditory and somatosensory stimuli shape activity, both individually and when combined, in the modular and matrix regions of the LC has remained unknown.

The major obstacle to progress has been the location of the LC on the lateral edge of the inferior colliculus where it cannot be accessed *in vivo* using conventional imaging approaches. The authors overcame this obstacle by developing methods to implant a microprism adjacent to the LC. By redirecting light from the lateral surface of the LC to the dorsal surface of the microprism, the microprism enabled two-photon imaging of the LC via a dorsal approach in anesthetized and awake mice. Then, by crossing GAD-67-GFP mice with Thy1-jRGECO1a mice, the authors showed that they could identify LC modules *in vivo* using GFP fluorescence while assessing neural responses to auditory, somatosensory, and multimodal stimuli using Ca²⁺ imaging. Critically, the authors also validated the accuracy of the microprism technique by directly comparing results obtained with a microprism to data collected using conventional imaging of the dorsal-most LC modules, which are directly visible on the dorsal IC surface, finding good correlations between the approaches.

Through this innovative combination of techniques, the authors found that matrix neurons were more sensitive to auditory stimuli than modular neurons, modular neurons were more sensitive to somatosensory stimuli than matrix neurons, and bimodal, auditory-somatosensory stimuli were more likely to suppress activity in matrix neurons and enhance activity in modular neurons. Interestingly, despite their higher sensitivity to somatosensory stimuli than matrix neurons, modular neurons in the anesthetized prep were overall more responsive to auditory stimuli than somatosensory stimuli (albeit with a tendency to have offset responses to sounds). This suggests that modular neurons should not be thought of as primarily representing somatosensory input, but rather as being more prone to having their auditory responses modified by somatosensory input. However, this trend was different in the awake prep, where modular neurons became more responsive to somatosensory stimuli. Thus, to this reviewer, one of the most intriguing results of the present study is the extent to which neural responses in the LC changed in the awake preparation. While this is not entirely unexpected, the magnitude and stimulus specificity of the changes caused by anesthesia highlight the extent to which higher-level sensory processing is affected by anesthesia and strongly suggests that future studies of LC function should be conducted in awake animals.

Together, the results of this study expand our understanding of the functional roles of matrix and module neurons by showing that responses in LC subregions are more complicated than might have been expected based on anatomy alone. The development of the microprism technique for imaging the LC will be a boon to the field, finally enabling much-needed studies of LC function *in vivo*. The experiments were well-designed and well-controlled, the limitations of two-photon imaging for tracking neural activity are acknowledged, and appropriate statistical tests were used.

<https://doi.org/10.7554/eLife.93063.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

This study provides important new insights into how multisensory information is processed in the lateral cortex of the inferior colliculus, a poorly understood part of the auditory midbrain. By developing new imaging techniques that provide the first optical

access to the lateral cortex in a living animal, the authors provide convincing *in vivo* evidence that this region contains separate subregions that can be distinguished by their sensory inputs and neurochemical profiles, as suggested by previous anatomical and *in vitro* studies. Additional information and analyses are needed, however, to allow readers to fully appreciate what was done, and the comparison of multisensory interactions between awake and anesthetized mice would benefit from being explored in more detail.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

*In this paper, the authors provide a characterisation of auditory responses (tones, noise, and amplitude-modulated sounds) and bimodal (somatosensory-auditory) responses and interactions in the higher-order lateral cortex (LC) of the inferior colliculus (IC) and compare these characteristics with the higher order dorsal cortex (DC) of the IC - in awake and anaesthetised mice. Dan Llano's group has previously identified gaba'ergic patches (modules) in the LC distinctly receiving inputs from somatosensory structures, surrounded by matrix regions receiving inputs from the auditory cortex. They here use 2P calcium imaging combined with an implanted prism to - for the first time - get functional optical access to these subregions (modules and matrix) in the lateral cortex of IC *in vivo*, in order to also characterise the functional difference in these subparts of LC. They find that both DC and LC of both awake and anaesthetised mice appear to be more responsive to more complex sounds (amplitude-modulated noise) compared to pure tones and that under anesthesia the matrix of LC is more modulated by specific frequency and temporal content compared to the gabaergic modules in LC. However, while both LC and DC appear to have low-frequency preferences, this preference for low frequencies is more pronounced in DC. Furthermore, in both awake and anesthetized mice, somatosensory inputs are capable of driving responses on their own in the modules of LC, but very little (possibly not at all) in the matrix. However, bimodal interactions may be different under awake and anesthesia in LC, which warrants deeper investigation by the authors: They find, under anesthesia, more bimodal enhancement in modules of LC compared to the matrix of LC and bimodal suppression dominating the matrix of LC. In contrast, under awake conditions bimodal enhancement is almost exclusively found in the matrix of LC, and bimodal suppression dominates both matrix and modules of LC.*

The paper provides new information about how subregions with different inputs and neurochemical profiles in the higher-order auditory midbrain process auditory and multisensory information, and is useful for the auditory and multisensory circuits neuroscience community.

Strengths:

The major strength of this study is undoubtedly the fact that the authors for the first time provide optical access to a subcortical region (the lateral cortex of the inferior colliculus (i.e. higher order auditory midbrain)) which we know (from previous work by the same group) have optically identifiable subdivisions with unique inputs and neurotransmitter release, and plays a central role in auditory and multisensory processing. A description of basic auditory and multisensory properties of this structure is therefore very useful for understanding auditory processing and multisensory interactions in subcortical circuits.

Weaknesses:

I have divided my comments about weaknesses and improvements into major and minor comments. All of which I believe are addressable by the reviewers to provide a more

clear picture of their characterisation of the higher-order auditory midbrain.

Major comment:

(1) The differences between multisensory interactions in LC in anaesthetised and awake preparations appear to be qualitatively different, though the authors claim they are similar (see also minor comment related to figure 10H for further explanation of what I mean). However, the findings in awake and anaesthetised conditions are summarised differently, and plotting of similar findings in the awake figures and anaesthetised figures are different - and different statistics are used for the same comparisons. This makes it very difficult to assess how multisensory integration in LC is different under awake and anaesthetised conditions. I suggest that the authors plot (and test with similar statistics) the summary plots in Figure 8 (i.e. Figure 8H-K) for awake data in Figure 10, and also make similar plots to Figures 10G-H for anaesthetised data. This will help the readers understand the differences between bimodal stimulation effects on awake and anaesthetised preparations - which in its current form, looks very distinct. In general, it is unclear to me why the awake data related to Figures 9 and 10 is presented in a different way for similar comparisons. Please streamline the presentation of results for anaesthetised and awake results to aid the comparison of results in different states, and explicitly state and discuss differences under awake and anaesthetised conditions.

We thank the reviewer for the valuable suggestion. We only highlighted the similarities between the data obtained from anesthetized and awake preparations to indicate the ability to reproduce the technique in awake animals for future assessment. Identifying those similarities between the two experimental setups was based on the comparison between modules vs matrix or LC vs DC within each experimental setup (awake vs anesthetized). Therefore, the statistics were chosen differently for each setup based on the size of the subjects (n) within each experimental preparation. However, we agree with the reviewer's comment that there are differences between the anesthetized and awake data. To examine these differences, we ran the same statistics for Figure 5 (tonotopy of LC vs. DC-anesthetic animals) and Figure 9 (tonotopy of LC vs DC-awake animals). In addition, we added a new figure after Figure 9 to separate the statistical analysis from the maps. Accordingly, Figures 4 and 5 (maps and analysis, respectively -anesthetized animals) now match Figures 9 and 10 (maps and analysis, respectively – awake animals). We also did the same thing for Figures 7 (microprism imaging of the LC - anesthetized animals), 8 (imaging of the LC from the dorsal surface - anesthetized animals) as well as Figure 11 or old Figure 10 (microprism imaging of the LC - awake animals) to address the similarities and differences of the multisensory data between awake and anesthetized animals. We edited the text accordingly in the result and discussion sections.

(2) The claim about the degree of tonotopy in LC and DC should be aided by summary statistics to understand the degree to which tonotopy is actually present. For example, the authors could demonstrate that it is not possible/or is possible to predict above chance a cell's BF based on the group of other cells in the area. This will help understand to what degree the tonotopy is topographic vs salt and pepper. Also, it would be good to know if the gaba'ergic modules have a higher propensity of particular BFs or tonotopic structure compared to matrix regions in LC, and also if general tuning properties (e.g. tuning width) are different from the matrix cells and the ones in DC.

Thank you for the reviewer's suggestion. We have examined the tonotopy of LC and DC using two regression models (linear and quadratic polynomial) between the BFs of the cells and their location on the anatomical axis. Therefore, the tonotopy is indicated by a significant regression fit with a high R² between the BFs the cells, and their location within each structure. For the DC, there was a significant regression fit between the BFs of the cells and their locations over the rostromedial to the caudolateral axis. Additionally, the R² of the

quadratic polynomial fit was higher than that of the linear fit, which indicates a nonlinear distribution of cells based on their BFs, which is consistent with the presence of high-low-high tuning over the DC surface. Given that the microprism cannot image the whole area of the LC, and it images a slightly different area in each animal, it was very difficult to get a consistent map for the LC as well as a solid conclusion about the LC tonotopy. However, we have examined the regression fit between the BFs of cells and their location along the main four anatomical axes of the field of view obtained from each animal (dorsal to ventral), (rostral to caudal), (dorsocaudal to ventrorostral) (dorsorostral to ventrocoudal). Unlike the DC, the LC imaged via microprism showed a lower R2 for both linear and quadratic regression mostly in the dorsoventral axis. We show the fitting curves of these regressions in Figure 4-figure supplement 1 (anesthetized data) and Figure 9-figure supplement 1 (awake data). Despite the inconsistent tonotopy of the LC imaged via microprism, the modules were found to have a higher BFs median at 10 kHz compared to matrix that had a lower BFs median at 7.1 kHz, which was consistent across the anesthetized and awake animals. We have added these results in the corresponding spot in the results section (lines 193-197 and 361-364). We have examined the tuning width using the binarized receptive field sum (RFS) method in which each neuron was given a value of 1 if it responds to a single frequency (Narrow RF), but this value increases if the neuron responds to more neighbor frequencies (wide RF). We did this calculation across all the sound levels. Both DC and LC of the anesthetized animals had higher RFS mean and median than those of awake animals given that ketamine was known to broaden the RF. However, in both preparations (anesthetized and awake), the DC had a higher RFS mean than that of the LC, which could be consistent with the finding that the DC had a relatively lower SMI than the LC. To show these new data, we made a new Figure 10-figure supplement 1, and we edited the text accordingly [lines 372-379 & 527-531].

(3) Throughout the paper more information needs to be given about the number of cells, sessions, and animals used in each panel, and what level was used as n in the statistical tests. For example, in Figure 4 I can not tell if the 4 mice shown for LC imaging are the only 4 mice imaged, and used in the Figure 4E summary or if these are just examples. In general, throughout the paper, it is currently not possible to assess how many cells, sessions, and animals the data shown comes from.

Thank you for the reviewer's comment. We do apologize for not adding this information. We added all the information regarding the size of the statistical subjects (number of cells or number of animals used) for every test outcome. To keep the flow of the text, we added the details of the statistical tests in the legends of the figures.

(4) Throughout the paper, to better understand the summary maps and plots, it would be helpful to see example responses of the different components investigated. For example, given that module cells appear to have more auditory offset responses, it would be helpful to see what the bimodal, sound-only, and somatosensory responses look like in example cells in LC modules. This also goes for just general examples of what the responses to auditory and somatosensory inputs look like in DC vs LC. In general example plots of what the responses actually look like are needed to better understand what is being summarised.

Thank you for the reviewer's comment and suggestion. We modified Figure 6 and the text accordingly to include all the significant examples of cells discussed throughout the work.

Reviewer #2 (Public Review):

Summary:

The study describes differences in responses to sounds and whisker deflections as well as combinations of these stimuli in different neurochemically defined subsections of the lateral and dorsal cortex of the inferior colliculus in anesthetized and awake mice.

Strengths:

The main achievement of the work lies in obtaining the data in the first place as this required establishing and refining a challenging surgical procedure to insert a prism that enabled the authors to visualise the lateral surface of the inferior colliculus. Using this approach, the authors were then able to provide the first functional comparison of neural responses inside and outside of the GABA-rich modules of the lateral cortex. The strongest and most interesting aspects of the results, in my opinion, concern the interactions of auditory and somatosensory stimulation. For instance, the authors find that a) somatosensory-responses are strongest inside the modules and b) somatosensory-auditory suppression is stronger in the matrix than in the modules. This suggests that, while somatosensory inputs preferentially target the GABA-rich modules, they do not exclusively target GABAergic neurons within the modules (given that the authors record exclusively from excitatory neurons we wouldn't expect to see somatosensory responses if they targeted exclusively GABAergic neurons), and that the GABAergic neurons of the modules (consistent with previous work) preferentially impact neurons outside the modules, i.e. via long-range connections.

Weaknesses:

While the findings are of interest to the subfield they have only rather limited implications beyond it. The writing is not as precise as it could be. Consequently, the manuscript is unclear in some places. For instance, the text is somewhat confusing as to whether there is a difference in the pattern (modules vs matrix) of somatosensory-auditory suppression between anesthetized and awake animals. Furthermore, there are aspects of the results which are potentially very interesting but have not been explored. For example, there is a remarkable degree of clustering of response properties evident in many of the maps included in the paper. Taking Figure 7 for instance, rather than a salt and pepper organization we can see auditory responsive neurons clumped together and non-responsive neurons clumped together and in the panels below we can see off-responsive neurons forming clusters (although it is not easy to make out the magenta dots against the black background). This degree of clustering seems much stronger than expected and deserves further attention.

Thank you for the reviewer's comment. We do apologize if some areas in the manuscript were imprecisely written. For anesthetized and awake data, we have only emphasized the similarities between the two setups to show the ability to use micropism in awake animals for future assessment. To highlight the differences between anesthetized and awake animals, we have now run uniform statistics for all the data collected from both setups. Accordingly, we have edited Figures 4 and 5 (tonotopy-anesthetized) to match Figures 9 and new Figure 10 (tonotopy-awake). Also, we edited Figures 7 and 8 (multisensory- anesthetized) to match Figure 11 or old Figure 10 (multisensory- awake). We edited the text accordingly in the results section and discussed the possible differences between anesthetized and awake data in the discussion section [lines 521-553].

We agree with the reviewer's comment that the cells were topographically clustered based on their responses. Some of these clusters include the somatosensory responsive cells, which were located mostly in the modules (Figures 7D and 8E). Also, the auditory responsive cells with offset responses were clustered mostly in the modules (Figures 7C and 8F). Accordingly, we have edited the text to emphasize this finding.

We noticed also that some responsive cells to the tested stimulations were surrounded by nonresponsive cells. By comparing the response of the cells to different stimuli we found that while Figures 7 and 11 (old Figure 10) showed only the response of the cells to auditory stimulation (unmodulated broadband noise at 80 dB) and somatosensory stimulation (whisker deflection), some nonresponsive cells to these specific stimulations were found to be responsive to pure tones of different frequencies and amplitudes. As an indicator of the cells' viability, we additionally examined the spontaneous activity of the nonresponsive cells across different data sets. We note that spontaneous activity was rare for all cells even among the responsive cells to sound or somatosensory stimulations. This finding could be related to the possibility that the 2P imaging of calcium signals may not be sensitive enough to track spontaneous activity that may originate from single spikes. However, in some data sets, we have found that the cells that did not respond to any tested stimuli showed spontaneous activity when no stimulation was given indicating the viability of those cells. We have addressed the activity of the non-responsive cells in the text along with a new Figure 11-figure supplement 1.

We changed the magenta into a green color to be suitable for the dark background. Also, we have completely changed the color palette of all of our images to be suitable for color-blind readers as suggested by reviewer 1.

Reviewer #3 (Public Review):

The lateral cortex of the inferior colliculus (LC) is a region of the auditory midbrain noted for receiving both auditory and somatosensory input. Anatomical studies have established that somatosensory input primarily impinges on "modular" regions of the LC, which are characterized by high densities of GABAergic neurons, while auditory input is more prominent in the "matrix" regions that surround the modules. However, how auditory and somatosensory stimuli shape activity, both individually and when combined, in the modular and matrix regions of the LC has remained unknown.

The major obstacle to progress has been the location of the LC on the lateral edge of the inferior colliculus where it cannot be accessed in vivo using conventional imaging approaches. The authors overcame this obstacle by developing methods to implant a microp prism adjacent to the LC. By redirecting light from the lateral surface of the LC to the dorsal surface of the microp rism, the microp rism enabled two-photon imaging of the LC via a dorsal approach in anesthetized and awake mice. Then, by crossing GAD-67-GFP mice with Thy1-jRGECO1a mice, the authors showed that they could identify LC modules in vivo using GFP fluorescence while assessing neural responses to auditory, somatosensory, and multimodal stimuli using Ca²⁺ imaging. Critically, the authors also validated the accuracy of the microp rism technique by directly comparing results obtained with a microp rism to data collected using conventional imaging of the dorsal-most LC modules, which are directly visible on the dorsal IC surface, finding good correlations between the approaches.

Through this innovative combination of techniques, the authors found that matrix neurons were more sensitive to auditory stimuli than modular neurons, modular neurons were more sensitive to somatosensory stimuli than matrix neurons, and bimodal, auditory-somatosensory stimuli were more likely to suppress activity in matrix neurons and enhance activity in modular neurons. Interestingly, despite their higher sensitivity to somatosensory stimuli than matrix neurons, modular neurons in the anesthetized prep were far more responsive to auditory stimuli than somatosensory stimuli (albeit with a tendency to have offset responses to sounds). This suggests that modular neurons should not be thought of as primarily representing somatosensory input, but rather as being more prone to having their auditory responses modified by somatosensory input. However, this trend was reversed in the awake prep, where

modular neurons became more responsive to somatosensory stimuli than auditory stimuli. Thus, to this reviewer, the most intriguing result of the present study is the dramatic extent to which neural responses in the LC changed in the awake preparation. While this is not entirely unexpected, the magnitude and stimulus specificity of the changes caused by anesthesia highlight the extent to which higher-level sensory processing is affected by anesthesia and strongly suggest that future studies of LC function should be conducted in awake animals.

Together, the results of this study expand our understanding of the functional roles of matrix and module neurons by showing that responses in LC subregions are more complicated than might have been expected based on anatomy alone. The development of the microprism technique for imaging the LC will be a boon to the field, finally enabling much-needed studies of LC function in vivo. The experiments were well-designed and well-controlled, and the limitations of two-photon imaging for tracking neural activity are acknowledged. Appropriate statistical tests were used. There are three main issues the authors should address, but otherwise, this study represents an important advance in the field.

(1) Please address whether the Thy1 mouse evenly expresses jRGECO1a in all LC neurons. It is known that these mice express jRGECO1a in subsets of neurons in the cerebral cortex, and similar biases in the LC could have biased the results here.

Thank you for the reviewer's comment. In the work published by Dana, et al, the expression of jRGECO1a in all Thy1 mouse lines was determined by the brightness of the jRGECO1a in the soma. Given that some cells do not show a detected level of jRGECO1a fluorescence until activated, the difference in expression shown in different brain regions could be related to the level of neuronal activity at the time of sample processing and not the expression levels of the indicator itself. To the best of our knowledge, there is no antibody for jRGECO1a, which can be used for detecting the expression levels of the indicator regardless of the neuronal activity. To test the hypothesis that DC and LC have different levels of jRGECO1a, we examined the expression levels of jRGECO1a after we perfused the mice with high potassium saline to elicit a general neuronal depolarization in the whole brain. Then we immunostained against NeuN (the neuronal marker) to quantify the percentage of the neurons expressing jRGECO1a to the total number of neurons (indicated by NeuN). To have a fair comparison, we restricted our analysis to include the areas imaged only by 2P as some regions were not accessible by microprism such as the deep ventral regions of the LC. There is a similar % of cells expressing jRGECO1a in DC and LC. As expected, the neurons expressing jRGECO1a were only nonGABAergic cells. We addressed these findings in the new Figure 3-figure Supplement 1 as well as the corresponding text in the results [lines 178-184] and methods sections [lines 878-892].

(2) I suggest adding a paragraph or two to the discussion to address the large differences observed between the anesthetized and awake preparations. For example, somatosensory responses in the modules increased dramatically from 14.4% in the anesthetized prep to 63.6% in the awake prep. At the same time, auditory responses decreased from 52.1% to 22%. (Numbers for anesthetized prep include auditory responses and somatosensory + auditory responses.). In addition, the tonotopy of the DC shifted in the awake condition. These are intriguing changes that are not entirely expected from the switch to an awake prep and therefore warrant discussion.

Thank you for the reviewer's comment. To determine if differences exist between anesthetized and awake data, we have now used the same statistics and edited Figures 4,5,7,8,9, and 10 as well as added a new Figure 11. Accordingly, we have edited the result section and added a paragraph addressing the possible differences between the two preparations in the Discussion section [lines 521-553].

(3) For somatosensory stimuli, the authors used whisker deflection, but based on the anatomy, this is presumably not the only somatosensory stimulus that affects LC. The authors could help readers place the present results in a broader context by discussing how other somatosensory stimuli might come into play. For example, might a larger percentage of modular neurons be activated by somatosensory stimuli if more diverse stimuli were used?

We agree with the reviewer's point. Indeed, the modules are receiving different inputs from different somatosensory sources such as somatosensory cortex and dorsal column nuclei, which could indicate that the activity of the cells in the modular areas could be evoked by different types of somatosensory stimulations, which is an open area for future studies. We have discussed this point in the revised Discussion section [lines 516-520].

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Minor comments:

(1) Figure 3H: The lateral surface seems quite damaged by the prism. An example slice of the imaging area of each mouse would help the reader better understand the extent of damage the prism leaves in the area of interest.

Thank you for the reviewer's comment. We already have included such images in Figures 4A, 7A, and 9A to present the field of view of all prism experiments. However, we need to clarify the point of tissue damage. The insertion of microprism may be associated with some tissue damage as a result of making the pocket for the microprism to be inserted, but it is not possible to get neuronal signals from a damaged field of view. Therefore, we do not believe that there is tissue damage to the parts of the LC imaged by microprism. However, there may be some areas where the microprism is not in direct contact with the LC surface. These areas are located mostly in the periphery of the field of view, and they are completely black as they are out of focus (i.e., the left side of Figure 3B). The right side of Figure 3b as well as Figure 3A have some black areas, which present the vasculatures, where there are no red signals because of the lack of jRGECO1a expression in those areas.

(2) In relation to the data shown in Figure 4E it is claimed that LC is tuned to higher frequencies (lines 195-196). However, the majority of cells appear to be tuned to frequencies below 14kHz (with a median of 7.5 kHz), which is quite low for the mouse. I assume that the authors mean frequencies that are relatively higher than the DC, but it is worth mentioning in the text that the BFs found in the LC are quite low-frequency responses for the mouse.

Thank you for the reviewer's comment, which we agree with. We edited this part by acknowledging that around 50% of the LC cells had a low-frequency bias to 5 and 7.1 kHz. Then we mentioned that most of the LC cells are tuned to relatively higher frequencies than those of the DC [lines 215-218].

(3) Figure 5A-C: Is it the tone-responsive cells plus an additional ~22% of cells that respond to AM, or are there also cells that respond to tones that do not respond to AM. Please break down to which degree the tone and AM responsive cells are overlapping.

Thank you for the reviewer's comment and suggestion. We broke down the responsive cells into cells responsive only to pure tone (tone selective cells or Tone-sel) or to only AM-noise (noise selective cells or Noise-sel) as well as cells responding to both sounds (nonselective

cells or Non-sel). We examined the fractions of these categories of cells in both LC and DC within all responsive neurons. Accordingly, we have edited Figure 5A-C as well as the text [lines 229-243].

(4) Figure 5D. It is unclear to me how a cell is classified as SMI or TMI responsive after computing the SMI or TMI for each cell. What statistic was used to determine if the cell was responsive or not?

Thank you for the reviewer's comment. We do apologize for the confusion caused by Figures 5D and E. These figures do not show the values of SMI or TMI, respectively. Rather, the figures show the percentage of the spectrally or temporally modulated cells, respectively. At each sound level, the cells were categorized into two main types. The spectrally modulated cells are those responsive to pure tones or unmodulated noise, so they can detect the spectral features of the sound (old Figure 5D or new Figure 5E). The temporally modulated cells are those responsive to AM-noise, so they can detect the temporal features of the sound of complex spectra like the broadband noise (old Figure 5E or new Figure 5F). To clear this confusion, we removed the words SMI and TMI from the figures, and then we renamed the x-axis label into "% of spectrally modulated cells" and "% of temporally modulated cells" for Figures 5D (new 5E) and E (new 5F), respectively.

(5) Figure 5 D, E: Is the decrease in SMI and TMI modulated cells in the modules a result of simply lower sensitivity to sounds (i.e. higher response thresholds)? If a cell responds to neither tone, AM, or noise it will have a low SMI and TMI index. If this is the case that affects the interpretation, as it is then not a decrease in sensitivity to spectral or temporal modulation, but instead a difference in overall sound sensitivity.

Thank you for the reviewer's comment. We apologize for the confusion about Figures 5E and D, which did not show the SMI and TMI values. Rather, they show the percentage of spectrally or temporally modulated cells, respectively, as explained in our previous response. Therefore, Figure 5D shows the percentage of cells that can detect the spectral features of sound, while Figure 5E shows the percentage of cells that can detect the temporal features of sounds of complex spectra like broadband noise. Accordingly, Figures 5D and E show the sensitivity to different features of sound and not the overall sound sensitivity.

(6) Figure 7 and 8: What is the false positive rate expected of the responsive cells using the correlation cell flagging criteria? Especially given that the fraction of cells responsive to somatosensory stimulation in LC (matrix) is 0.88% and 1.3% in DC, it is important to know what the expected false positive rate is in order to be able to state that there are actually somatosensory responses there or if this is what you would expect from false positives given the inclusion test used. Please provide an estimate of the false positive rate given your inclusion test and show that the rate found is statistically significantly above that level - and show this rate with a line in Figure 7 H, I.

Thank you for the reviewer's comment. To test the efficiency of the correlation method to determine the responsive cells, we initially ran an ROC curve comparing the automated method to a blinded human interpretation. The AUC of the ROC curve was 0.88. This high AUC value indicates that the correlation method can rank the random responsive cells than the random nonresponsive cells. At the correlation coefficient (0.4), which was the cutoff value to determine the responsive cells for somatosensory stimulation, the specificity was 87% and the sensitivity 72%, the positive predictive value was 73%, and the negative predictive value was 86%. Although the above percentages indicate the efficiency of the correlation method, we excluded all the false responsive cells from the analysis. Therefore, the fractions of cells in the graphs are the true responsive cells with no contamination of the non-responsive cells. We also modified Figures 7H and I to match the other data sets obtained from awake animals.

Therefore, Figures 7H and I no longer show the average of the responsive cells. Instead, they show the % of different fractions of responsive cells within each cellular motif (modules and matrix). Accordingly, we believe that there is no need to include a rate line on the graph. We added the section describing the validation part to the methods section [lines 808-815].

(7) Figure 7: Please clarify what is meant by a cell responding to 'both responding to somatosensory and auditory stimulation'. Does it mean that the cell has responses to both auditory and somatosensory stimulation when presented individually or if it responds to both presented together? If it is the former, I don't understand how the number to both can be higher than the number of somatosensory alone (as both requires it also to respond to somatosensory alone). If it is the latter (combined auditory and somatosensory) then it seems that somatosensory inputs remove the responsiveness of most cells that were otherwise responsive to auditory alone (e.g. in the module while 42% respond to sound alone, combined stimulation would leave only 10% of cells responsive). Please clarify what exactly the authors are plotting and stating here.

Thank you for the reviewer's comment. The responsive cells in Figure 7 are divided into three categories. Each category has a completely different group of cells. The first category is for the cells responding only to auditory stimulation (auditory-selective cells or Aud-sel). The second category is for the cells that respond only to somatosensory stimulation (somatosensory selective cells or Som-sel). The third category is for the cells that respond to both auditory and somatosensory stimulations when both stimulations are presented individually (auditory/somatosensory nonselective cells or Aud/Som-nonsel). Accordingly, the number of cells may be different across all these categories. We have clarified this part in the text [lines 299-303]. We have modified Figures 7, 8, and 11 (old Figure 10) to match the data from anesthetized and awake animals, so Figures 7H and I now show the collective % of the cells from all animals within modules vs matrix.

(8) Why are the inferential statistics used in Figure 9F (chi-square test) and Figure 5A-C (t-test) when it tests the same thing (the only difference is one is anaesthetised data and the other awake)? Indeed, all Figure 9 and 10 (awake data figures) plots use chi-square tests to test differences in percentages instead of t-tests used in earlier (anaesthetised data figures) plots to test differences in percentages between groups. Please clarify the reason for this change in statistics used for similar comparisons.

Thank you for the reviewer's comment. Imaging the LC via microprism from awake animals confirmed the ability to run this technique with no interference to the ambulatory functions of the animals. Therefore, the main goal was to highlight the similarities between the data obtained from awake and anesthetized setups by highlighting the comparison between the LC and DC or between modules and matrix within each preparation (anesthetized vs awake). Accordingly, the statistics used to run these comparisons were chosen based on the number of the tested animals at each setup (7 anesthetized animals and 3 awake animals for prism insertion). The low number of animals used for awake data made us use the number of cells collectively from all animals instead of the number of animals, so we used the Chi-square test to examine the differences in percentages.

(9) Figure 10H: The main text describes the results shown here as similar to what was seen in anaesthetised animals. But it looks to me like the results in awake animals are qualitatively different from the multisensory interaction seen in anaesthetised animals. In anaesthetised animals the authors find that there is a higher chance of auditory responses being enhanced by somatosensory inputs when cells are in the modules compared to in the matrix. However, in awake data, this relationship is flipped, with more bimodal enhancement found in the matrix compared to the modules. Furthermore, almost all cells in the modules are suppressed by combined somatosensory input which

looks like it is different from what is found in anaesthetised mice and what is described in the discussion: 'we observed that combined auditory-somatosensory stimulation generally suppressed neural responses to auditory stimuli and that this suppression was most prominent in the LC matrix'.

Thank you for the reviewer's comment. Our statement was meant to show how the data obtained from awake and anesthetized animals were generally similar. However, we agree that the statement may not be suitable due to the possible differences between awake and anesthetized animals. To address a fair comparison between the anesthetized and awake preparations, we ran similar statistics and graphs for Figures 7, 8, and 11 (old Figure 10). Given that the areas occupied by modules and matrix are different across animals due to the irregular shape of the modules, we chose to run a chi-square test for all the data to quantify the collective % of responding cells within modules vs matrix from all tested animals for each experimental setup (anesthetized vs awake). The anesthetized and awake animals similarly showed that modules and matrix had higher fractions of auditory responsive cells. However, matrix had more cells responding to auditory stimulations than modules, while modules had more cells responding to somatosensory stimulation than matrix. In contrast, while the anesthetized animals showed higher fractions of offset auditory-responsive cells, which were mostly clustered in the modules, the offset auditory-responsive cells were very rare in awake animals (6 cells/one animal).

Based on the fractions of cells with suppressed or enhanced auditory response induced by bimodal stimulation, the data obtained from anesthetized and awake animals showed that the auditory response in the matrix was suppressed more than enhanced by bimodal stimulation. In contrast, modules had different profiles across the experimental setups and locations. For instance, the modules imaged via microprism in the anesthetized and awake animals showed suppressed more than enhanced auditory responses, but modules imaged from the dorsal surface in anesthetized animals showed enhanced more than suppressed auditory responses. Additionally, modules had less suppressed and more enhanced auditory responses compared to matrix in the anesthetized animals regardless of the location of the modules (microprism or dorsal surface). Yet, modules from awake animals had more suppressed and less enhanced auditory responses compared to matrix. We have addressed these differences in the results and discussion section.

Additional minor comments that I think the authors could use to aid their manuscript clarity:

(1) The figure colour selection - especially in Figures 7 and 8 - is really hard to tell apart. Please choose more distinct colours, and a colour scheme that is appropriate for colour blind readers.

Thank you for the reviewer's suggestion. We have noticed that the magenta color assigned for the cells with offset responses was very difficult to distinguish from the black background. We have changed the magenta color to green to be different from the color of other cells. Using Photoshop, we chose a color scheme that is suitable for color-blind readers in all our maps.

(2) The sentence in lines 331-334 should be rephrased for clarity.

Thank you for the reviewer's suggestion. We have rephrased the statement for clarity [lines 364-371].

Reviewer #2 (Recommendations For The Authors):

As mentioned in the public review the strong clustering evident in some of the maps (some of which may be related to module/matrix differences but certainly not all of it) seems worth scrutinizing further. Would we expect such a strong spatial segregation of auditory responsive and non-responsive neurons? Would we expect response properties (e.g. off-responsiveness) other than frequency tuning to show evidence of a topographic arrangement in the IC? In addressing this it would, of course, be important to rule out that this clustering is not down to some trivial experimental variables and truly reflects functional organization. For instance, are the patches of non-responsive neurons found in parts of the field of view with poor visibility, poor labelling, etc which may explain why it is difficult to pick up responses there? Are the neurons in non-responsive areas otherwise active (i.e. do they show spontaneous activity) or could they be 'dead'? Could the way neuropil signals are dealt with play a role here (it is weighted by 0.4 which strikes me as quite low)? In relation to this, I am also wondering to what extent the extreme overrepresentation (Figure 4) of neurons with a BF of 5kHz (some of this is, of course, down to the fact that the lower end of the frequency range was 5kHz and that the step size was 0.5 octaves), especially in the DC, is to be interpreted.

Thank you for the reviewer's comment. Before analysis, the ROIs of all cells were set around the cell bodies using the jRGECO1a signals as a reference, so all cells (responsive and nonresponsive) were collected from areas of good visibility of jRGECO1a signals. In other words, no cells were collected from regions having poor jRGECO1a signals. In Figures 7, 8, and 11 (old Figure 10), the cells showed response either only to unmodulated broadband noise at 80 dB as an auditory stimulus or to whisker deflection with specific speed and power as a somatosensory stimulus. Given that the two stimuli above had specific parameters, the remaining non-responsive cells may respond to auditory or somatosensory stimulations with other features. For instance, some nonresponsive cells to the unmodulated broadband noise were responding to pure tones with different amplitudes and frequencies or to different AM-noise with different amplitudes and modulation frequencies. Also, these nonresponsive cells may not respond to any of our tested stimuli and may respond to other sensory stimulations. Some of the non-responsive cells showed spontaneous activity when no stimulations were presented. However, we can not rule out the possibility that some of these nonresponsive cells may not be viable. We have addressed the clustering properties in the revised version of the manuscript in the corresponding spots of the results and discussion sections. We have added a new supplementary figure (Figure 11- Figure Supplement 1) to show how the nonresponsive cells to the unmodulated noise may respond to other types of sound and to show the spontaneous activity of some non-responsive cells.

For the neuropil, previous reports used the contamination factor (r) in a range of 0.3-0.7 (we referenced these studies in the method section [line 776] based on the tissue or cells imaged, vasculatures, and the objective used for imaging. Therefore, we optimized the contamination factor (r) to be 0.4 through a preliminary analysis based on the tissue we image (LC), and the objective used (16x with NA = 0.8 and 3 mm as a working distance).

We agree that there is an overrepresentation of 5 kHz as the best tuning frequency for DC cells. The previous report (A. B. Wong & Borst, 2019) showed a large zone of the DC where cells were tuned to (2-8 kHz). Given that 5kHz was the lowest tested frequency in our experiment, we think that the low-frequency bias of the DC surface is consistent between studies. This finding also could be supported by the electrophysiology data obtained by spanning the recording electrodes through the IC tissue along the dorsoventral axis. In those experiments, the cells were tuned to lower frequencies at the dorsal surface of the IC.

We have changed the magenta-colored cells to green ones, so it will be easier to identify the cells. As required by another reviewer, we changed the color pallets of some images and

cellular maps to be suitable for color-blind readers.

The manuscript would benefit from more precise language in a number of places, especially in the results section.

Line 220/221, for instance: "... a significant fraction of cells that did not respond to pure tones did respond to AM-noise" Strictly speaking, this sentence suggests that you considered here only the subset of neurons that did not respond to pure tones and then ran a test on that subset. The test that was done seems to suggest though that the authors tested whether the percentage of responsive cells was greater for pure tones or for AM noise.

Thank you for the reviewer's comment. We do apologize for the confusion. In the revised manuscript, we categorized the cells according to their response into cells responding to pure tone only (tone-selective cells or Tone-sel), Am-noise only (noise-selective cells or Nose-sel), and to both pure tone and am-noise (nonselective cells or Non-sel). We have modified Figure 5 accordingly. We did the same thing for the data obtained from awake animals and showed that in a new figure to easily match the analysis done for the anesthetized animals.

Please refer to the figure panels in the text in consecutive order. 2B, for instance, is mentioned after 2H.

Thank you for the reviewer's comment. Throughout the paper, we kept the consecutive order of the figure panels within each figure to be in a smooth flow with the text. Yet, figure 2 was just the only exception for a good reason. Figure 2 is a complex one that includes many panels to show a parallel comparison between LC imaged via microprism and DC through single photon images, two-photon images, validating laser lesioning, and histology. Accordingly, we navigated many panels of the figure to efficiently highlight the aspects of this comparison. We prefer to keep Figure 2 as one figure with its current format to show this parallel comparison between LC and DC.

The legend for Figure 2 could be clearer. For instance, there are two descriptions for panel D. Line 1009: "(C-E)" [i.e. C, D, E] and line 1010: "(D and F)".

Thank you for the reviewer's comment. It should be C and E, not C-E. We have fixed the mistake [line 1224]

Line 275: What does 'with no preference' mean?

Thank you for the reviewer's comment. We do apologize for the confusion. There are three categories of cells. Some cells respond only to auditory stimulation, while others respond to only somatosensory stimulation. However, there is another group of cells that respond nonselectively to auditory and somatosensory stimulations or Aud/Som-nonsel cells. We edited the sentence to be clearer [lines 303-304].

Line 281 (and other places): What does 'normalized against modules' mean?

Thank you for the reviewer's comment. This normalization was done by dividing the number of responsive cells of the same response type in the matrix by that in the modules. Therefore, the value taken by modules was always 1 and the value taken by the matrix is something around 1. Accordingly, the value for matrix could be > 1 if matrix had more cells than modules. In contrast, the value of matrix would be < 1 if matrix had fewer cells than modules. In the revised version, we used this normalization method to make the revised Figures 5C

and 10C to describe the cell fractions responding to pure tone only, AM-noise only, or to both stimuli in the matrix vs modules.

Sentence starting on line 288. I don't find that point to be as obvious from the figures as the sentences seem to suggest. Are we to compare magenta points (auditory off cells) from 7C with green points in 7F?

Thank you for the reviewer's comment. We came to this conclusion based on our visual comparison of magenta points (now green in the revised version to increase the visibility) representing the auditory offset cells in Figure 7C and the green points in Figure 7F representing the cells responding to both somatosensory and auditory stimulations. In the revised manuscript, we statistically examined if the percentage of onset auditory response and offset auditory responses are different within the responsive cells to both somatosensory and auditory stimulations in the modules vs matrix. We have found that most of the cells responding to both somatosensory and auditory stimulations inside the modules had offset auditory responses, which could indicate a level of multisensory integration between somatosensory input and the offset auditory responses in these cells. We have added the statistical results to the revised manuscript to address this effect [lines 312-317]

Lines 300-302: "These data suggest that the module/matrix system permits preservation of distinct multimodal response properties in the face of massive integration of inputs in the LC". First, I'm not quite sure what that sentence means. Second, it would be more appropriate for the discussion. Third, the fact that we are more likely to find response enhancement in the modules than in the matrix is nicely consistent with the idea (supported by work from the senior author's lab and others) that excitatory somatosensory input predominantly targets neurons in the modules (which is why we see mostly response enhancement in the modules) and that this input targets GABAergic neurons which then project to and inhibit neurons both outside and inside of their module. Therefore, I would recommend that the authors replace the aforementioned sentence with one that interprets these results in light of what we know about this somatosensory-auditory circuitry.

Thank you for the reviewer's comment. Despite the massive multimodal inputs, the LC receives from auditory vs nonauditory regions, the module/matrix system is a platform for distinct multimodal responses indicated by more somatosensory responsive cells in modules versus more auditory responsive cells in matrix, which matches the anatomical differences that were reported before. We edited the sentence in the light of the comparison between the data obtained from awake and anesthetized animals and moved it to the discussion section [lines 503-506].

The term 'LC imaged via microprism' is used dozens of times throughout the manuscript. Replacing it with a suitable acronym or initialism could improve the flow of the text and would make some of the sentences less cumbersome.

Thank you for the reviewer's suggestion. We changed the term "LC imaged via microprism" into LC (microprism) throughout the revised manuscript.

5A-C: It is unclear what is being compared here. What are the Ns? Different animals?

Thank you for the reviewer's comment. We do apologize for this missing information. We have added the number of subjects used in every statistical test in each corresponding figure legend.

5G: minus symbol missing on the y-axis.

Thank you for the reviewer's comment. We gladly have fixed that.

Figure 6: Are these examples or population averages?

Thank you for the reviewer's question. Every figure panel of the old Figure 6 represents a single trace of an example cell. However, we modified Figure 6 to include more examples of cells showing different responses complying with another reviewer's suggestion. Each panel of the new Figure 6 represents the average response of 5 stimulations of the corresponding stimulus type. We preferred to show the average signal because it was the one used for the subsequent analysis.

How are module borders defined?

Thank you for the reviewer's question. The modules were defined based on the intensity of the green channel that shows the expression of the GFP signals. The boundaries of modules were determined according to the distinction between high and low GFP signal boundaries of the modules. This step was done before data analysis to avoid any bias.

7JKL: How are these to be interpreted? Does panel 7K, for instance, indicate that the fraction of neurons showing 'on' responses was roughly twice as large in the matrix than in the modules and that the fraction of neurons showing 'off' responses was roughly 10 times larger in the modules than in the matrix (the mean seems to be at about 1/10).

Thank you for the reviewer's comment. The data represented by Figures 7J-L defined the normalization of the number of cells of the same response type in the matrix against the modules. This normalization was done per animal, and then the data of the matrix were plotted against the normalization line at 1 representing the modules. The matrix will be claimed to have more cells than modules if the median of the matrix values > 1 . In contrast, the matrix will be claimed to have fewer cells than the modules if the median of the matrix values < 1 . Finally, if the median of matrix values $= 1$, this means there is no difference between matrix and modules. However, to match the data obtained from anesthetized animals (Figures 7 and 8) with those obtained from awake animals (Figure 11 or old Figure 10), we ran all data through the Chi-square test in the revised manuscript. Therefore, the format of Figures 7K-L was changed in the revised manuscript, so they became new Figures 7I-K.

10A suggests that significantly more than half the neurons shown here are not auditory responsive. Perhaps I am misinterpreting something here but isn't that in contrast to what is shown in panel 9F?

Thank you for the reviewer's comment. The data shown in Figure 10A (or revised Figure 11A) represents the cellular response to only one stimulus (broadband noise at 80 dB with no modulation frequency), while Figure 9F (revised 10B) represents the cells responding to varieties of auditory stimulations of different combinations of frequencies and amplitudes (pure tones) as well as to AM-noise of different amplitudes and modulation frequencies. Accordingly, the old Figure 9F or revised Figure 10B shows different cell types based on their responses. For instance, some cells respond only to pure tone. Others respond only to AM-noise or to both pure tones and AM-noise. This may also support that the nonresponsive cells in Figure 10A (revised 11A) can respond to other types of sound features.

The way I understood panels 7L and 8K there were more suppressed neurons in the matrix than in the modules (line 296: "cells in the modules had a higher odds of having an enhancement response to bimodal stimulation than matrix, while cells in the matrix had a higher odds of having a suppressive response to bimodal stimulation"). Now, panel 10F

indicates that in awake mice there is a greater proportion of suppressed neurons in the modules than in the matrix. I may very well have overlooked or misread something but I may not be the only reader confused by this so please clarify.

Thank you for the reviewer's comment. We do apologize for this confusion. The ambiguity between Figures 7 and 8 (anesthetized animals) as well as Figure 10 (awake animals) comes from the fact that different statistics have been used for each preparation. In the revised version, we have fixed that by running the same statistics for all the data, and we accordingly revised Figures 7, 8, and 10 (new Figure 11). In brief, the matrix preserves a higher percentage of cells with suppressed auditory responses than those with enhanced auditory responses induced by bimodal stimulation in all conditions (anesthetized vs awake). In contrast, modules act differently across all tested conditions. While modules had more cells with enhanced auditory responses induced by bimodal interaction in anesthetized animals, they had more cells with suppressed response in awake animals indicating that modules could be sensitive to the effect of anesthesia compared to matrix. We addressed this effect in the discussion of the revised manuscript [lines 521-553].

Line 438: ...as early AS...

Thank you for the reviewer's comment. We gladly fixed that [line 512].

Reviewer #3 (Recommendations For The Authors):

My minor recommendations for the authors are as follows:

(1) The text can be a bit difficult to follow in places. This is partly due to the convoluted nature of the results, but I suggest a careful read-through to look for opportunities to improve the prose. In particular, there is a tendency to use long sentences and long paragraphs. For example, the third paragraph of the introduction runs for almost fifty lines.

Thank you for the reviewer's comment and suggestion. We have fixed that.

(2) This might be due to journal compression, but some of the bar graphs in the figures are difficult to read. For example, the individual data points, especially when filled with striped background colors get lost. Axes can become invisible, like the y-axis in 7L, and portions of bars, like in 7F, are not always rendered correctly. Error bars are sometimes hidden behind data points, as in 5C. Increasing line thickness and shifting individual data points away from error bars might help with this.

Thank you for the reviewer's comment and suggestion. We made all the data points with black color and filled circles to make the data points visible. We put all the data points behind the main columns, so they don't block the error bars. We have fixed figures 7 and 5.

(3) Throughout the manuscript, the authors use a higher SMI to indicate a preference of cells for auditory stimuli with "greater spectral... complexity" (e.g., lines 219 and 372). I find this interpretation a bit challenging since SMI compares a neuron's preference for tones over noise, and to me, tones seem like the least spectrally complex of all auditory stimuli. Perhaps some clarification of what the authors mean by this would help. For example, is the assumption that a neuron that prefers tones over noise is, either directly or indirectly, receiving input sculpted by inhibitory processes?

Thank you for the reviewer's comment. In general, higher SMI values indicate an increase in the preference of the cells to respond to pure tones than noise with no modulation (less spectral complexity). We will clarify this statement throughout the manuscript. However, the

SMI value was not mentioned in lines 219 and 372. The statement mentioned in line 219 describes the revised figure 5C (old 5B), where more cells in matrix specifically respond to AM-noise compared to modules, which indicates the preference of the matrix to respond to sounds of greater spectral and temporal complexity. The statement in 372 in the discussion section refers to the finding in revised figures 5E and F (old 5D and E). In the revised figure 5E or old 5D, the data show that matrix has more cells responding to pure tones or noise with no modulation than modules, so matrix has a lower threshold to detect the spectral features of sound (revised figure 5E or old 5D). In the revised figure 5F or old 5E, the data show that matrix has more cells responding to AM-noise than modules, which indicates that matrix functions more to process the temporal features of sound. As explained above, all findings were related to the percentage of cells responding to specific sound stimuli and not the exact SMI values. We have revised the figures accordingly by removing the terms SMI and TMI from the figures, and we have clarified that in the text.

(4) Lines 250-253: How does a decrease in SMI correspond to "an increase in pure tone responsiveness?" Doesn't a decrease suggest the opposite?

Thank you for the reviewer's comment, which we agree with. We do apologize for that. We have fixed this statement [lines 275-277] and any related findings accordingly.

(5) Line 304: Add "imaged via microprism" or similar after "response profiles with the LC."

Thank you for the reviewer's suggestion. We have fixed that. However, we changed the term "LC imaged via microprism" into "LC(microprism)" for simplicity as suggested by another reviewer [line 330].

(6) Figure 5A and C: Both plots show that more neurons responded to AM-noise than tones, but it would be interesting to know how much the tone-responsive and AM-noise responsive populations overlapped. Were all tone-responsive neurons also responsive to AM-noise?

Thank you for the reviewer's comment. We have categorized the cells based on their response to pure tone only, AM-only, and both pure tone and AM-noise when each stimulus is presented individually. We have modified Figures 5A and C, and they are now Figures 5B and D.

(7) Figure 5G: Missing negative sign before "0.5."

Thank you for the reviewer's suggestion. We gladly have fixed that. However, old Figure 5G became a revised Figure 5H.

(8) Figure 7 legend, Line 1102: Missing period after "(C and E)".

Thank you for the reviewer's suggestion. We think that the period should be placed before (C and E) at the end of "respectively". The parentheses refer to the statements after them. We gladly fixed that. [line 1394]

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